



Robotic Systems

Chapter VII Motion and Path Planning

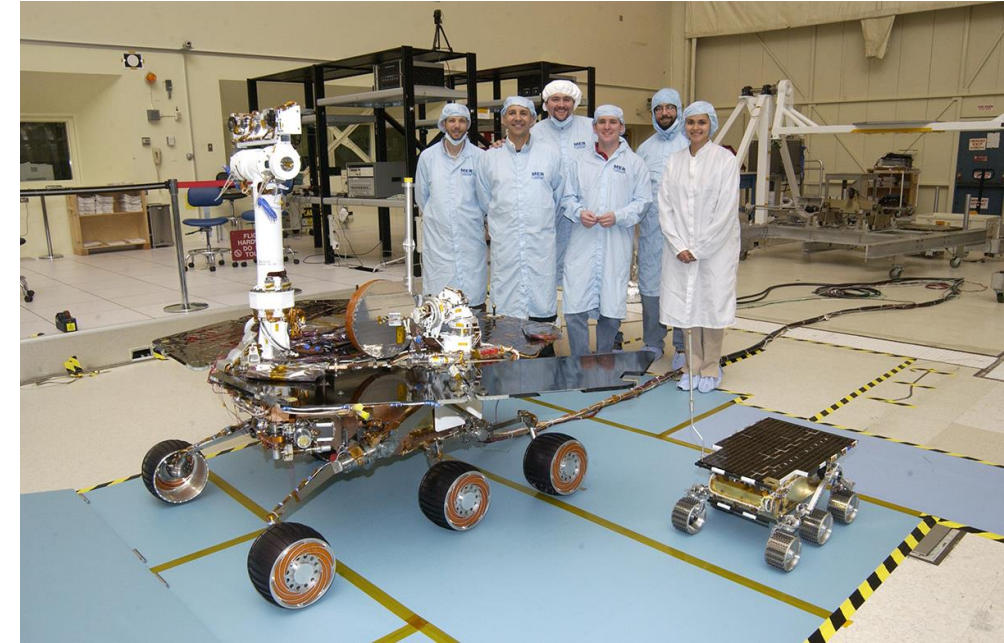
Prof. Dr. Ibrahim Hamarash
University of Kurdistan-Hewlêr

Mobile robots:

A class of robots that are able to move through the environment.



Examples of mobile robots in the field



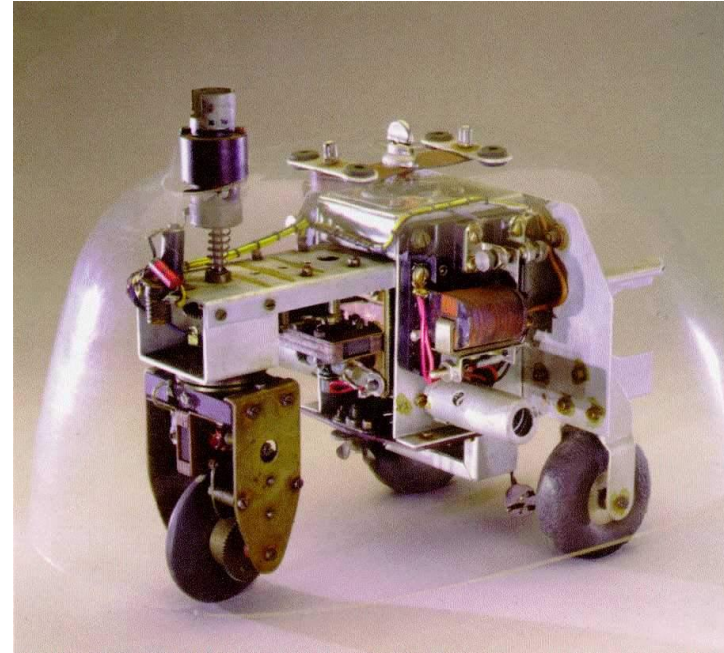
REMARK

However mobile robots are very similar in terms of **what they do and how they do it**. One of the most important functions of a mobile robot is to **move to some place**. That place might be specified in terms of some feature in the environment, for instance move to the **light**, or in terms of some **geometric coordinate or map reference**. In either case the robot will take some path to reach its destination and it faces challenges such as **obstacles** that might block its way or having an **incomplete map**, or **no map at all**.

Path building strategies

One strategy is to have very simple sensing of the world and to react to what is sensed. For example **Elsie the robotic tortoise** was built in the 1940s and **reacted** to her environment to seek out a light source without having any explicit plan or knowledge of the position of the light.

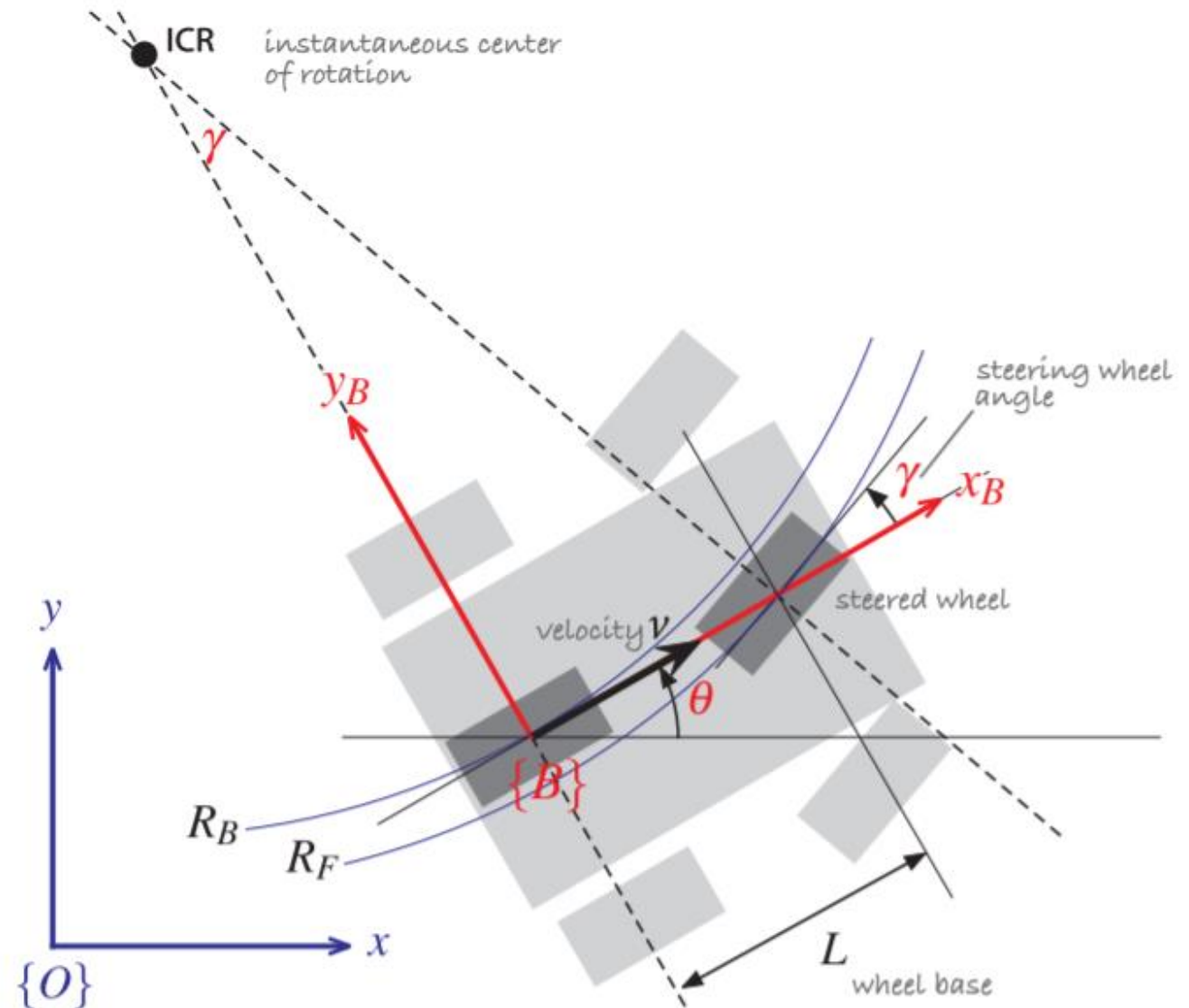
An alternative to the reactive approach was embodied in the 1960s robot Shakey, which was capable of 3D perception and created a map of its environment and then reasoned about the map to plan a path to its destination.



Modelling Car-Like Mobile Robots

Cars with steerable wheels are a very effective class of vehicle and the archetype for most ground robots

The car is shown in light grey, and the bicycle approximation is dark grey. The vehicle's body frame is shown in red, and the world (Reference) coordinate frame in blue. The **steering wheel angle** is γ and the velocity of the back wheel, in the x-direction, is v . The two wheel axes are extended as dashed lines and intersect at the **Instantaneous Center of Rotation (ICR)** and the distance from the ICR to the back and front wheels is R_B and R_F respectively.

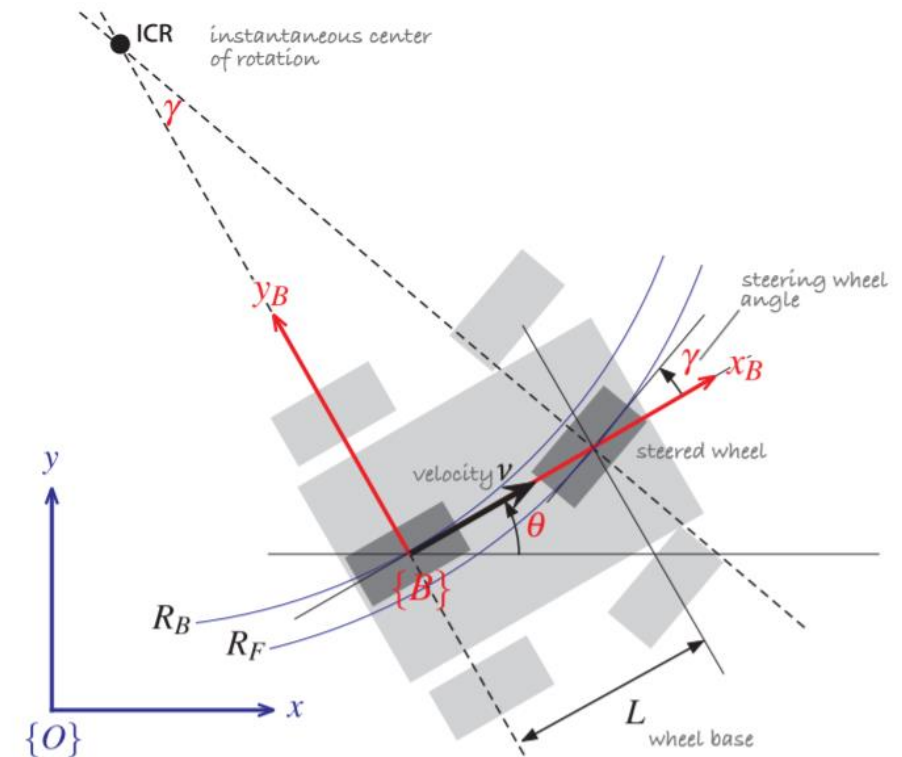


Modelling Car-Like Mobile Robots, cont.

Assume that the velocity of each wheel is in the plane of the wheel, and that the wheel rolls without slipping sideways

$${}^B\mathbf{v} = (v, 0)$$

The pose of the vehicle is represented by its body coordinate frame $\{B\}$ with its x-axis in the vehicle's forward direction and its origin at the center of the rear axle. The configuration of the vehicle is represented by the generalized coordinates $q = (x, y, \theta) \in C$ where $C \subset \mathbb{R}^2 \times S$.



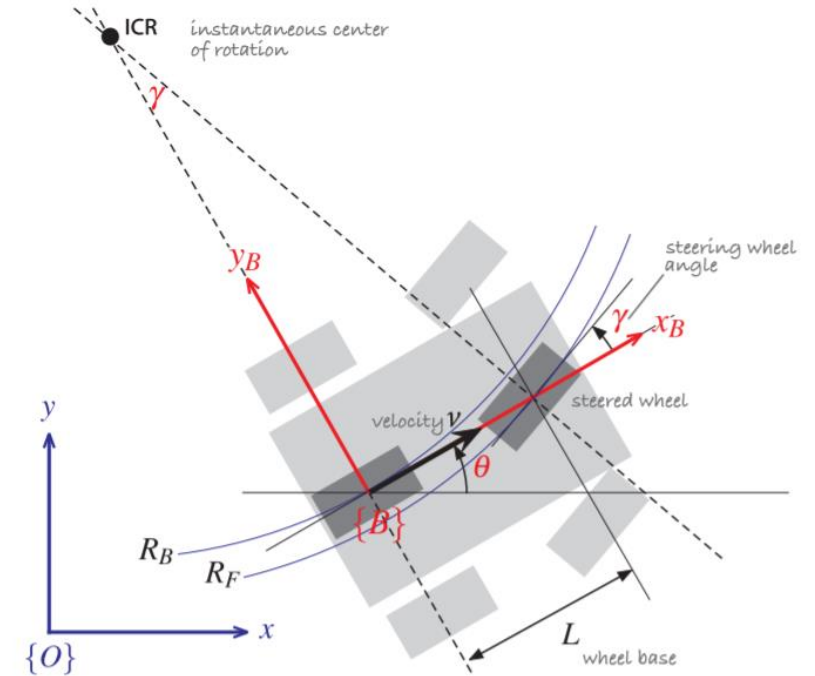
Modelling Car-Like Mobile Robots, cont.

The dashed lines show the direction along which the wheels cannot move, the lines of no motion, and these intersect at a point known as the Instantaneous Center of Rotation (ICR). The reference point of the vehicle thus follows a circular path and its angular velocity is

$$\dot{\theta} = \frac{v}{R_B}$$

The velocity of the robot in the world frame is $(v \cos \theta, v \sin \theta)$ and combined with the angular velocity, the equations of motion be,

$$\begin{aligned}\dot{x} &= v \cos \theta \\ \dot{y} &= v \sin \theta \\ \dot{\theta} &= \frac{v}{L} \tan \gamma\end{aligned}$$

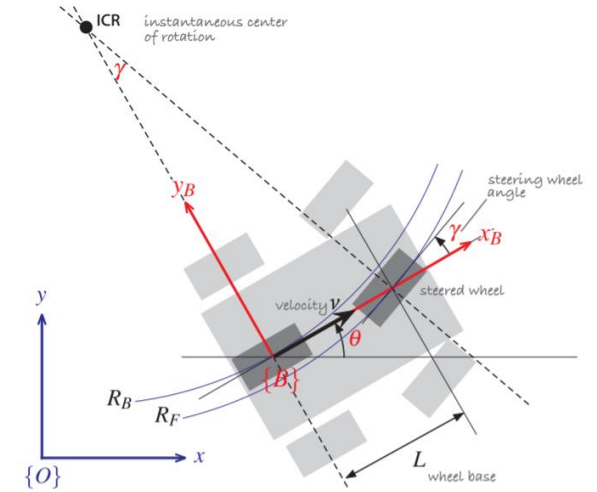
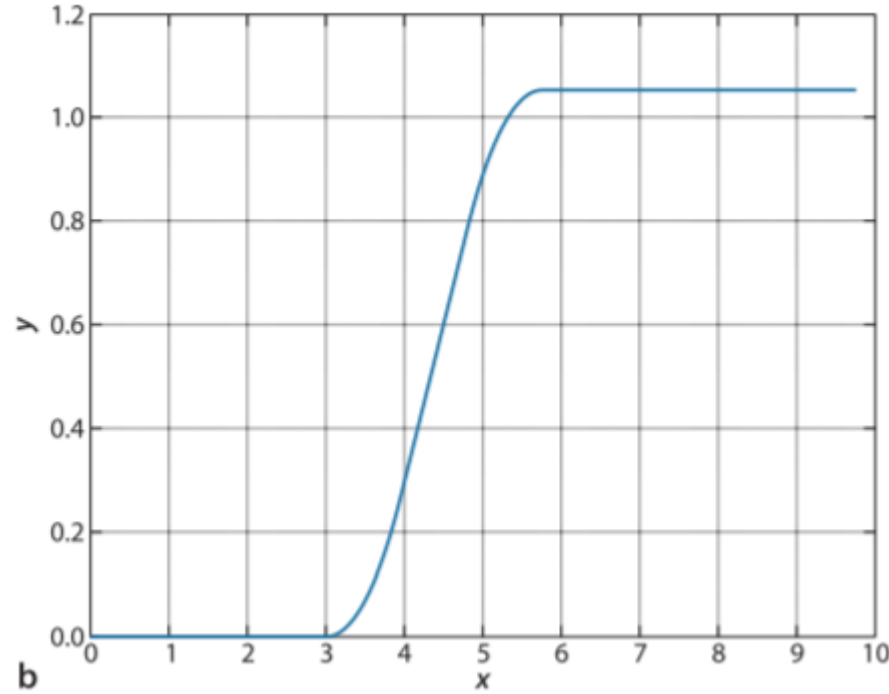
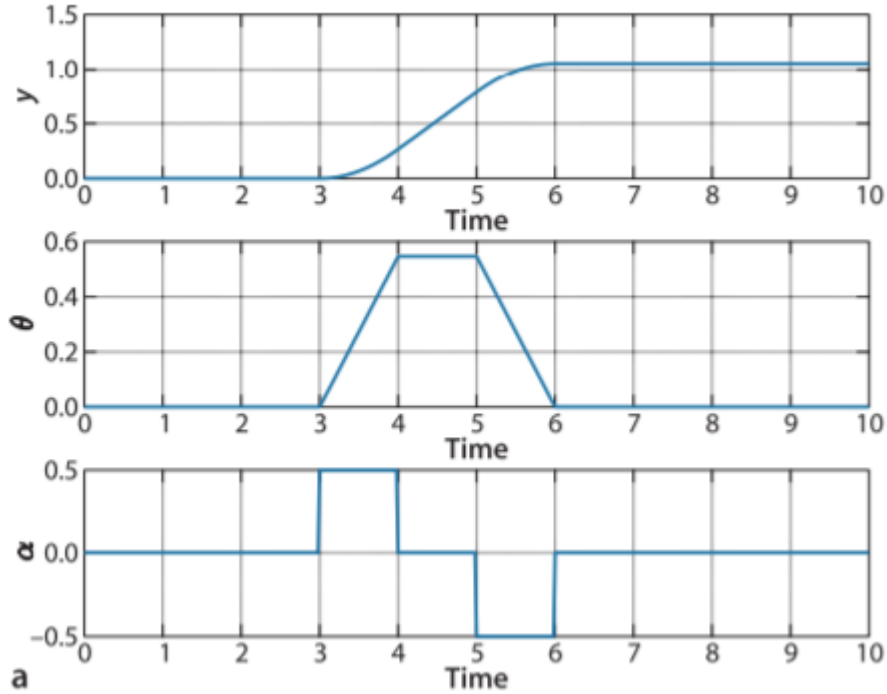


REMARK

- This model is referred to as a kinematic model since it describes the velocities of the vehicle but not the **forces** or **torques** that cause the velocity.
- The rate of change of heading $d/dt(\theta)$ is referred to as **turn rate**, **heading rate** or **yaw rate** and can be measured by a gyroscope.
- The motion equation captures some other important characteristics of a car-like vehicle. When $v = 0$ then $d/dt(\theta) = 0$; that is, it is not possible to change the vehicle's orientation when it is not moving. As we know from driving, we must be moving in order to turn. When the steering angle $\gamma = \pi/2$ the front wheel is orthogonal to the back wheel, the vehicle cannot move forward and the model enters an undefined region.

Motion Analysis

The velocity input is a constant, and the steering wheel angle is a finite positive pulse followed by a negative pulse.



Example

Moving to a point

Consider the problem of moving toward a goal point (x^*, y^*) in the plane. We will control the robot's velocity to be proportional to its distance from the goal,

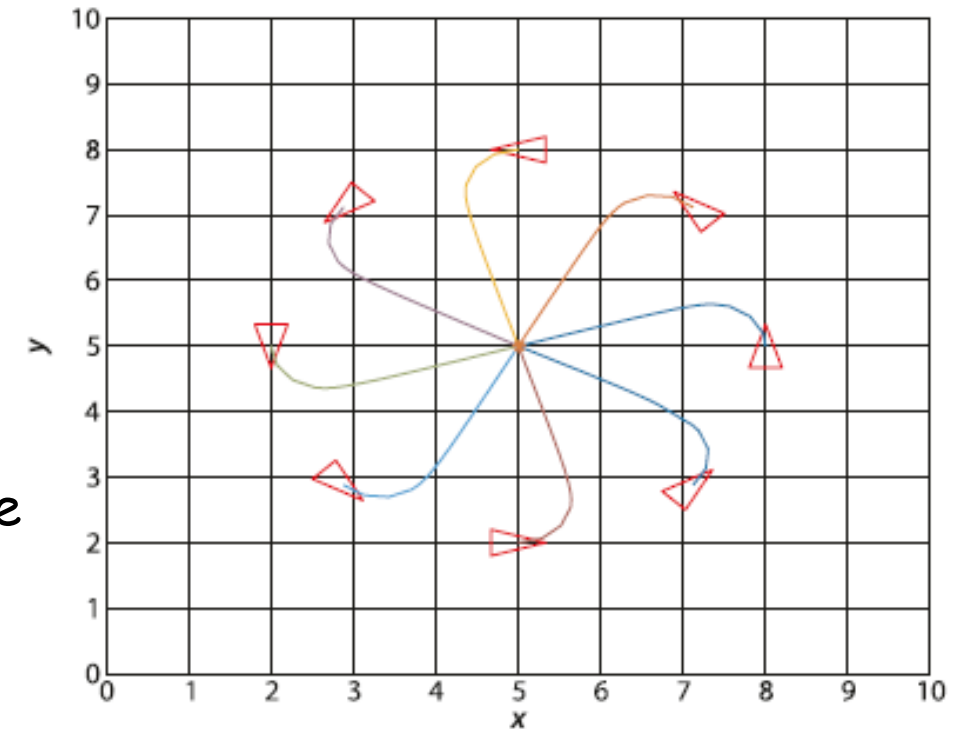
$$v^* = K_v \sqrt{(x^* - x)^2 + (y^* - y)^2}$$

and to steer toward the goal which is at the vehicle-relative angle in the world frame of

$$\theta^* = \tan^{-1} \frac{y^* - y}{x^* - x}$$

using a proportional controller

$$\gamma = K_h (\theta^* \ominus \theta), \quad K_h > 0$$



Example Following a line

Another useful task for a mobile robot is to follow a line on the plane defined by $ax+by+c=0$. This requires two controllers to adjust steering.

One controller

$$\alpha_d = -K_d d, K_d > 0$$

turns the robot toward the line to minimize the robot's normal distance from the line,

$$d = \frac{(a, b, c) \cdot (x, y, 1)}{\sqrt{a^2 + b^2}}$$

Example Following a line, cont.

The second controller adjusts the heading angle, or orientation, of the vehicle to be parallel to the line

$$\theta^* = \tan^{-1} \frac{-a}{b}$$

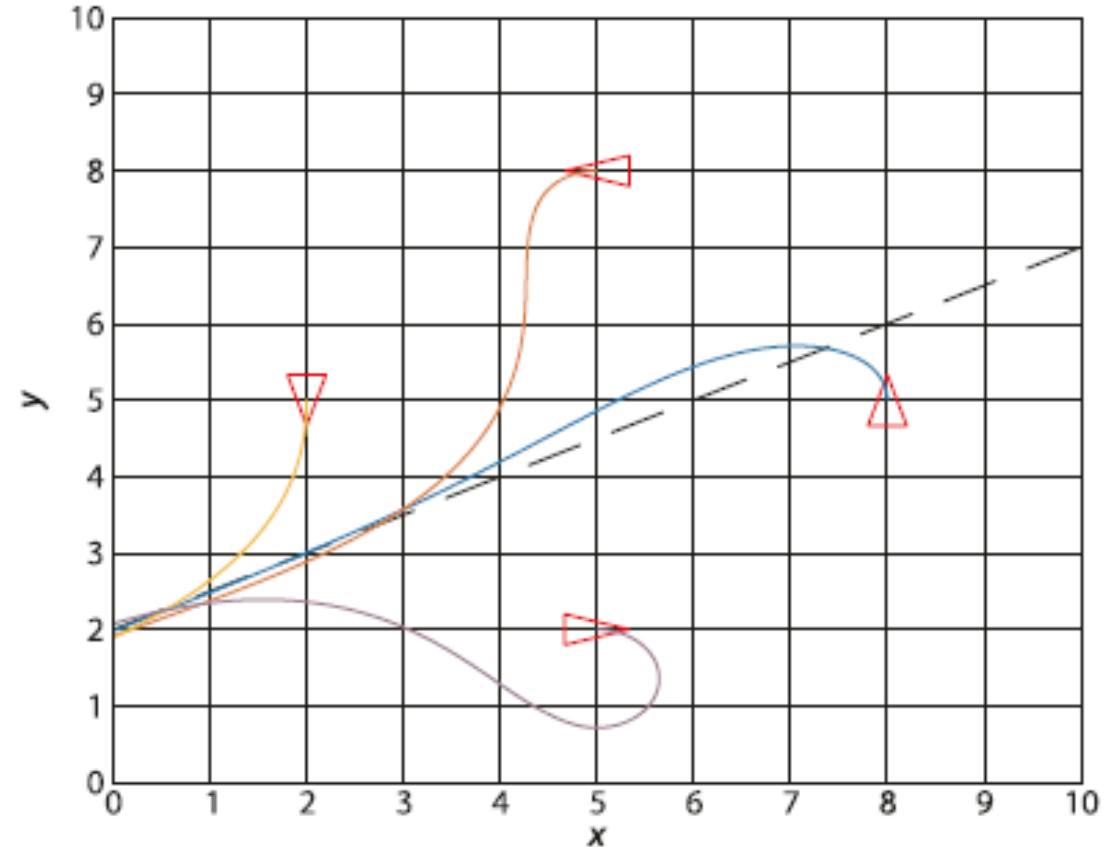
using the proportional controller

$$\alpha_h = K_h (\theta^* \ominus \theta), K_h > 0$$

The combined control law

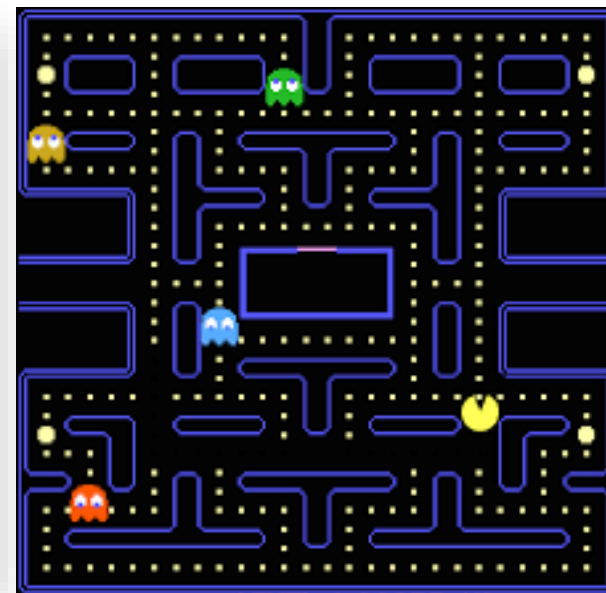
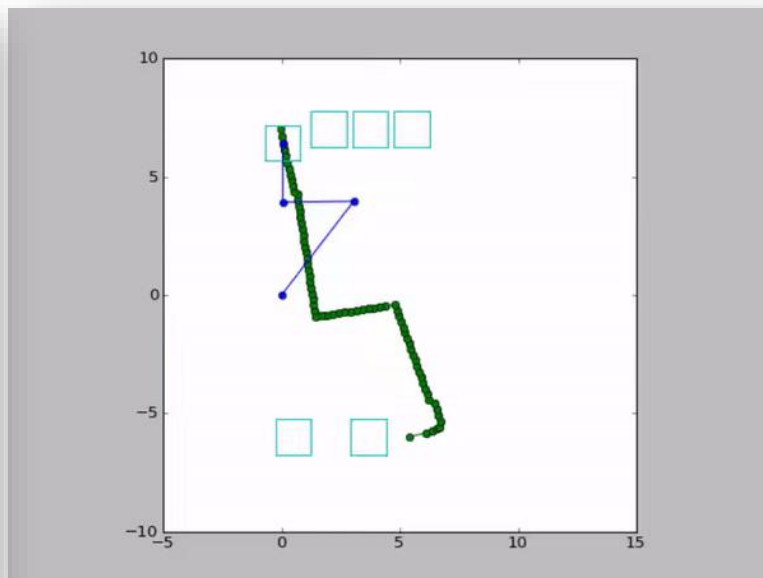
$$\gamma = -K_d d + K_h (\theta^* \ominus \theta)$$

turns the steering wheel so as to drive the robot toward the line and move along it.



Motion Planning is the process of breaking down a desired movement task into motions satisfying given constraints (as not hitting obstacles, keeping speed limits) and possibly address optimality aspects.

Determining where to move without hitting obstacles.



Motion Planning types

Robots come in a variety of sizes, types, applications and control modes, but despite all their diversity, robots share navigation and motion planning as a fundamental characteristic which is the major issue in the robotic industry.

Based on motion, robots are classified into manipulators with fixed location movable arms and mobile robotics with displacement capabilities.



Trajectory and Path planning

The robotics' Motion contains two parts; **trajectory** and **path planning**.

- A trajectory is defined as "generating the reference inputs for the robot control system that ensures the implementation of the desired motion". It is about introducing the proper **force** and **torque** to the joints and compliant at the end effector.
- Path planning, on the other hand, is defined as "finding a collision-free path from an initial (source) to a final configuration (target)". It is about the generation of a geometric path between the start and target positions taking into considerations optimisation and obstacle avoidance.

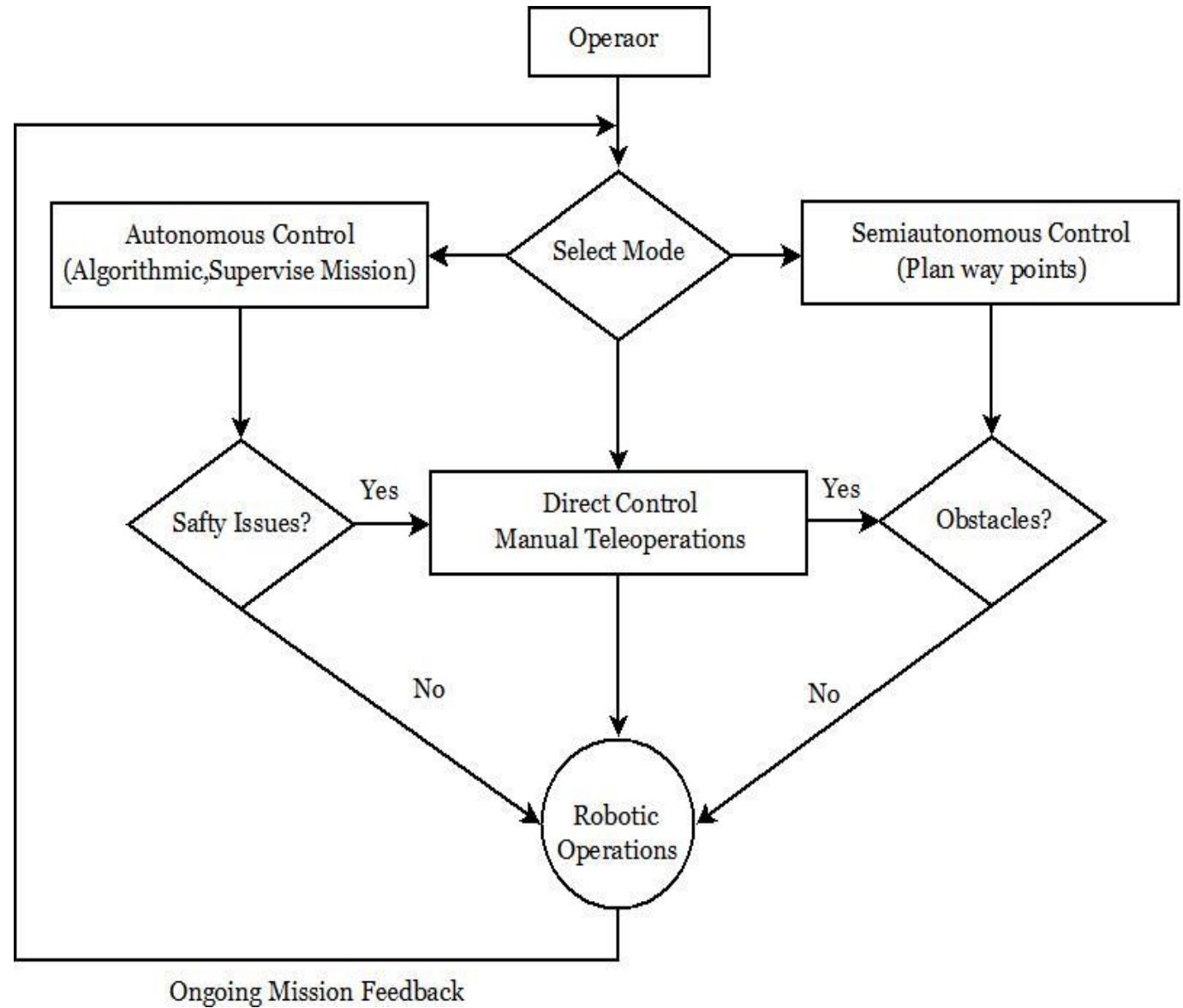
Path:

- Consists of ordered loci of points in the space which the robot should follow.
- The path provides a pure geometric description of motion
- The path is usually planned globally taking into account obstacle avoidance.

Trajectory:

- It is the path plus velocities and accelerations in its each point. The trajectory is specified and designed locally.
- Parts of the path are covered by individual trajectories.

Motion Control Modes



General Motion Planning Purpose

- The goal of a general purpose planning problem is to come up with a sequence of actions that accomplish a given goal.
- In many cases, this boils down to a kind of geometry problem. Where the goal is to guide the robot through a particular trajectory that avoids all the obstacles that we know about in the environment.
- This basic approach can be applied to a wide variety of robotic systems including relatively simple robots that roll around on the ground, to robotic arms with multiple degrees of freedom

Path Planning Objective

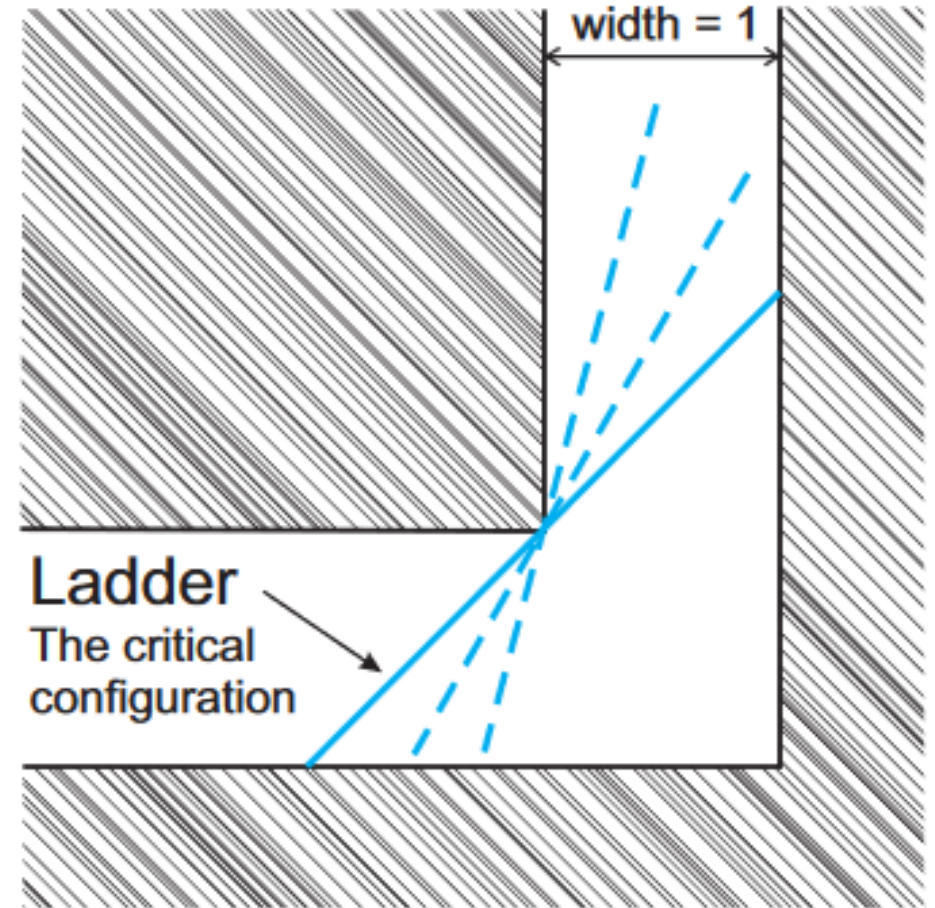
The path planning problem of robotics technology is a **geometric problem**. It is defined as the finding of a collision-free optimal path from a start point (initial) to an endpoint (target) in the environment (workspace, $\mathcal{W}$) with uncertainties to a group for autonomous robots.

Motion Planning and Computational Geometry

The geometric aspect of the task (spatial reasoning) induces use of methods from computational geometry.

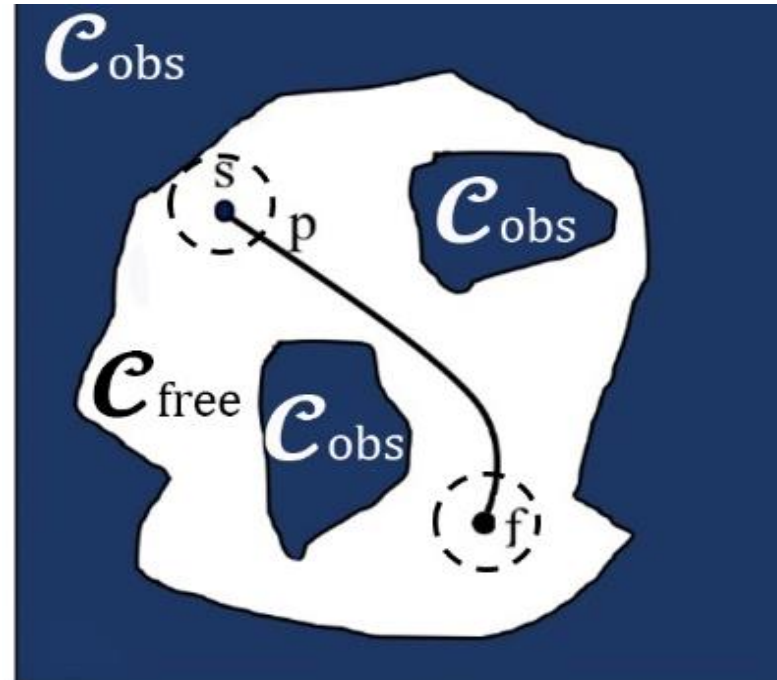
Example: What is the longest ladder that can be moved around a right-angled corridor of unit width?

For a straight rigid ladder, the answer is $2\sqrt{2}$, which allows the ladder to just pivot around the corner at a 45° angle



Path Planning Modeling

Let $\mathcal{R}$ be a robot with rigid sub-parts moving in Euclidean space (The workspace) $\mathcal{W}$, $\mathcal{W} \subset \mathcal{R}^d$, $d=2$ or 3 for planar or spatial spaces respectively. In an obstacle exitance space, $\mathcal{W}$ consists of two regions; a free space $\mathcal{C}_{\text{free}} \subset \mathcal{R}^d$ and obstacle space $\mathcal{C}_{\text{obs}} \subset \mathcal{R}^d$, $\mathcal{W} = \mathcal{C}_{\text{free}} \cup \mathcal{C}_{\text{obs}}$



Path Planning Model, cont.

“Given a space $\mathcal{W}$ contains obstacles space $\mathcal{C}_{\text{obs}}$, find the paths $p_i \subset \mathcal{C}_{\text{free}} \mid \mathcal{C}_{\text{obs}}$ such that p_i connects any two points $s_n, f_m \cup \mathcal{C}_{\text{free}} \mid \mathcal{C}_{\text{obs}}$ and $p_i \cup \mathcal{C}_{\text{free}} \mid p_j$ $i \neq j$, minimizing the measure J ”.

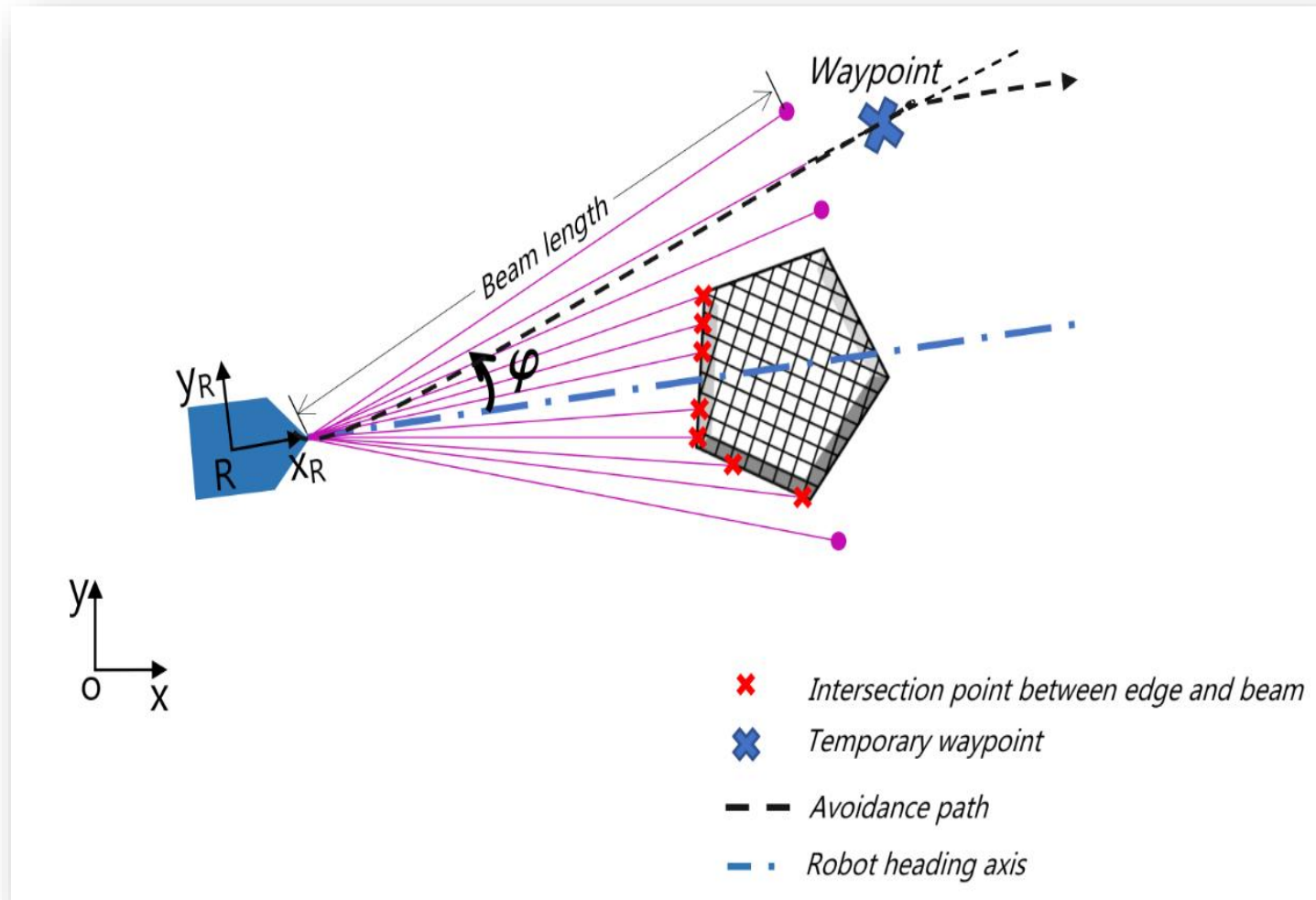
Where s is the start position and f is the final target, J is a cost measure to be minimised, P is the connected path between s and f , n is the initial and m is the final pose respectively.

Obstacle Avoidance Algorithms

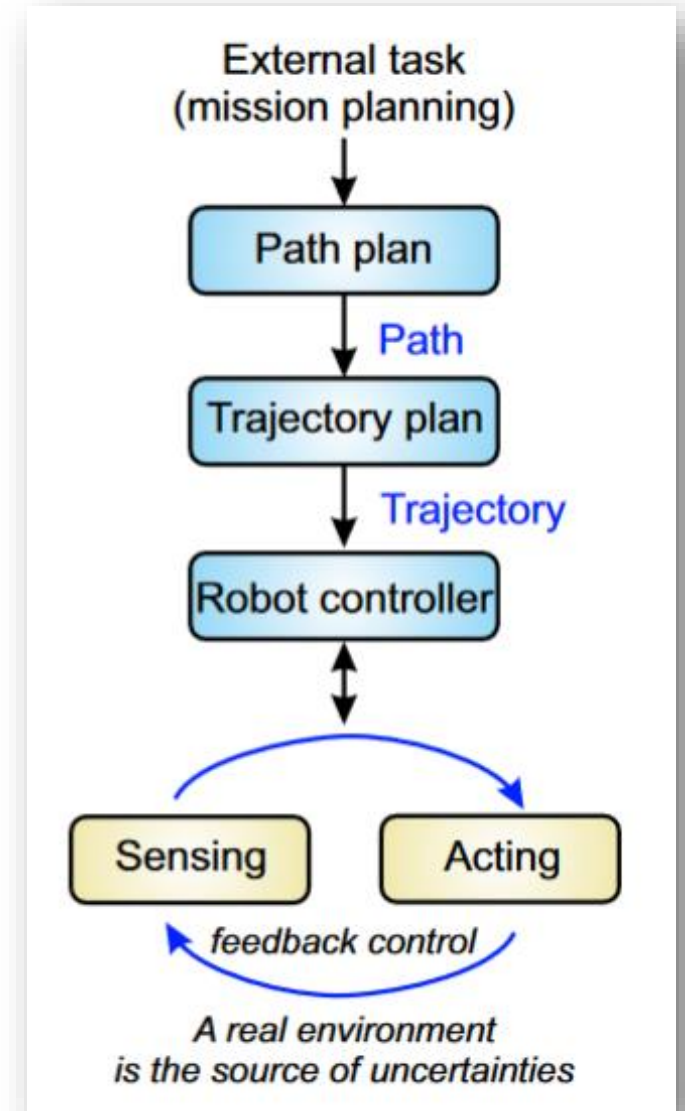
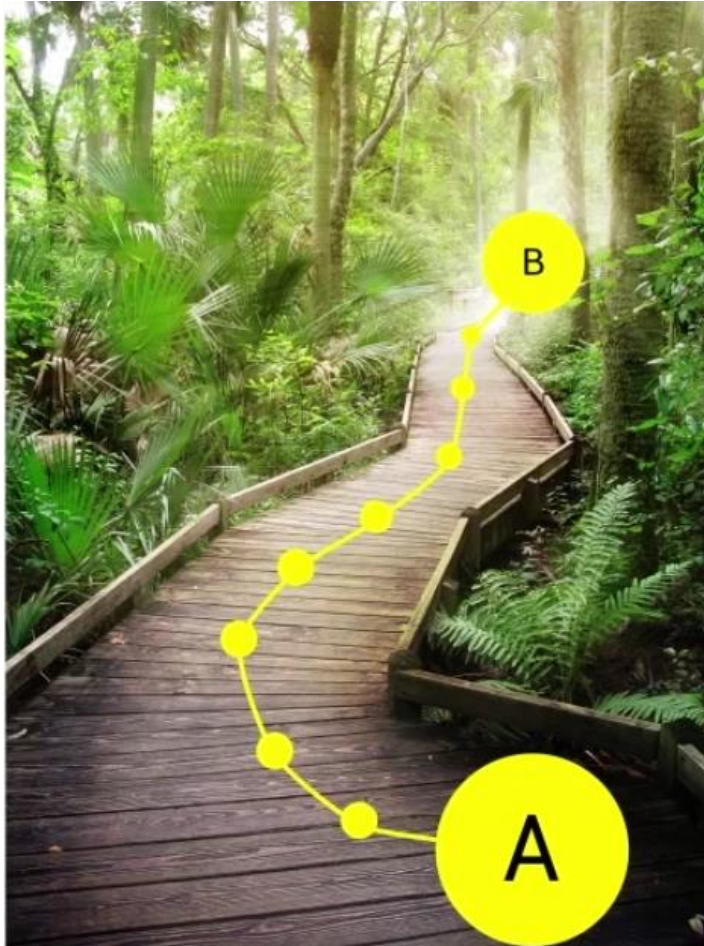
Example:

Given an obstacle r and configuration of a robot q , let the Euclidean distance between the obstacle side and the robot centre in the heading direction be $d(r,q)$, then the following criteria are applied as an obstacle detection criteria:

- $d(r,q) > 0$ There is no contact with the obstacle
- $d(r,q) = 0$ There is contact with the obstacle
- $d(r,q) < 0$ Penetration of the obstacle



Example:



Motion Planning Case Study:

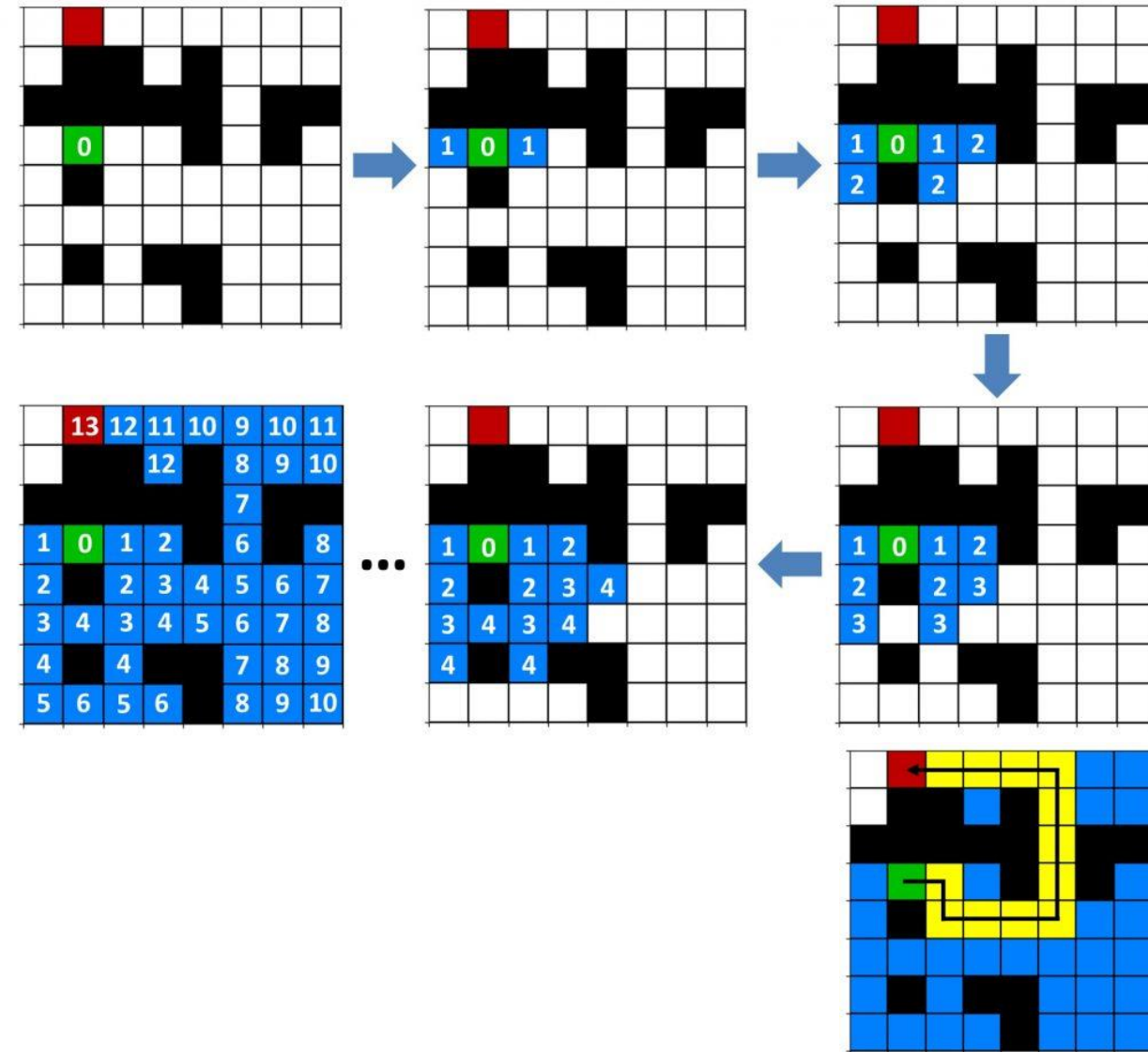
- Consider for example the problem that we would be facing if we were trying to design a robot that was supposed to do our laundry.
- We would love for that robot to be able to come up with the entire sequence of actions that needed to be performed on it's own. From finding laundry, taking it downstairs, putting it in the washing machine, taking it out; folding it; etc., etc.
- The Motion Planning Problem is actually more specifically concerned with coming up with plans to move a robot from one location to another. To get it from Point A to Point B.

Graph based algorithms for path planning

The goal is to come up with a procedure that the computer can use to guide the robot from its start location to its end location, while minimizing the total number of steps that are performed.

Mobile Robots, path planning algorithms: Graph-based algorithm

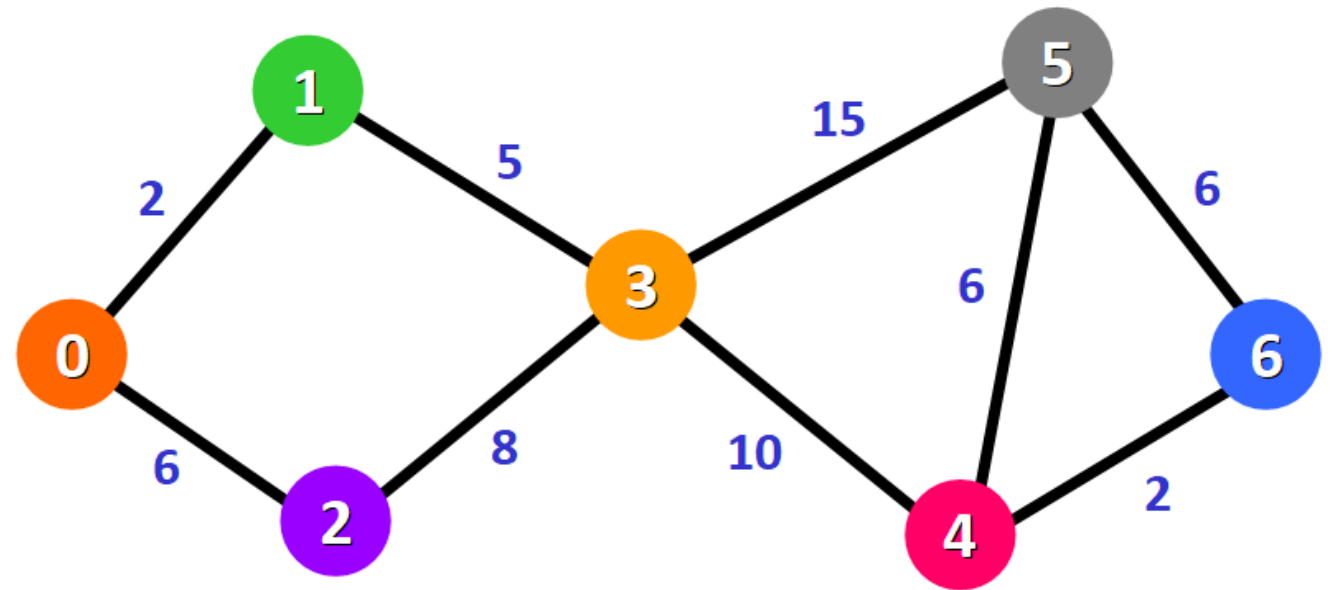
In graph theory, a graph is defined as a structure consisting of "**nodes**," i.e., points, and "**edges**," i.e., lines that connect the nodes. In graph-based path-planning, we travel from node to node along the edges, with the aim of figuring out how to get from some initial node to a destination node.



Dijkstra Algorithm

<https://www.freecodecamp.org/news/dijkstras-shortest-path-algorithm-visual-introduction/>

A* Algorithm

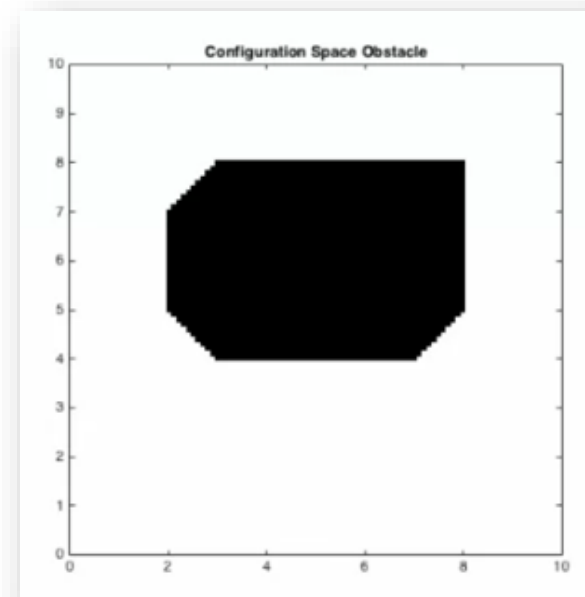
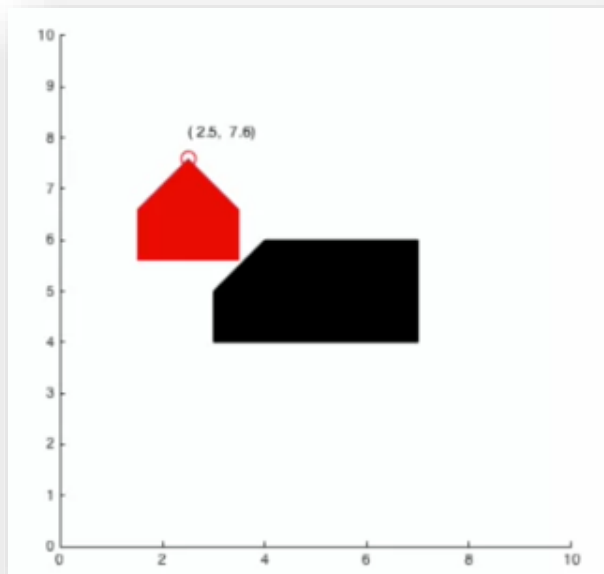
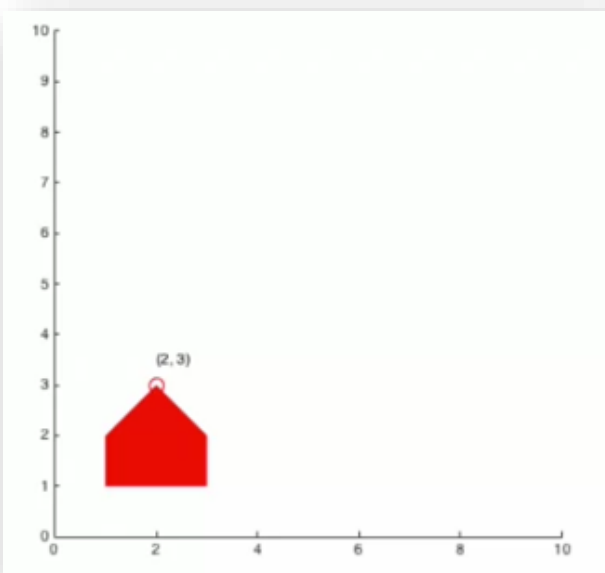


Configuration Space based motion planning

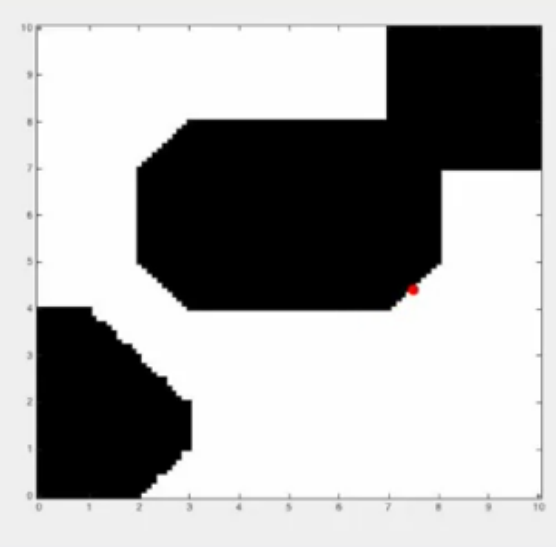
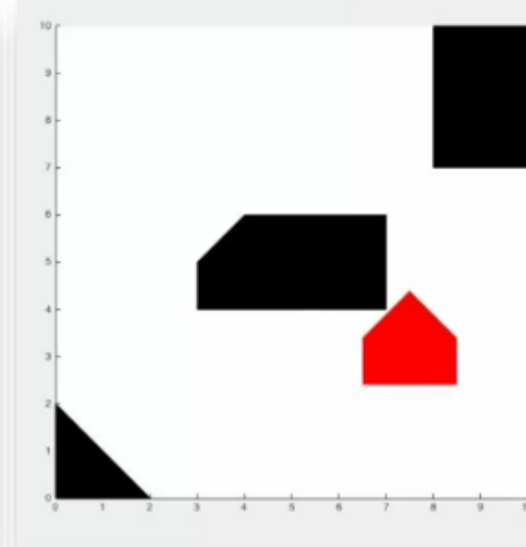
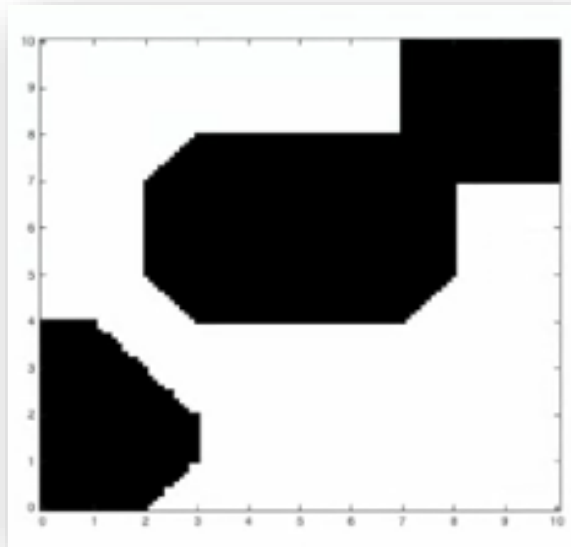
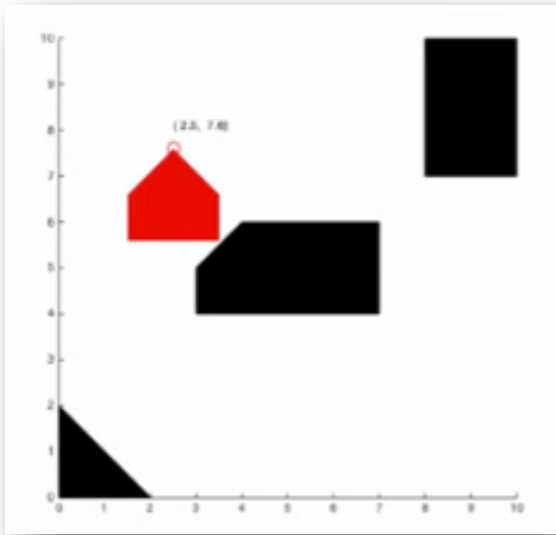
- In the motion planning problems we've basically reduced the problem to planning on a graph, where the robot can take on various discrete positions, which we can enumerate and connect by edges.
- In the real world, most of the robots we are going to build can move continuously through space.
- Configuration space is a handy mathematical and conceptual tool, which was developed to help us think about these kinds of problems in a unified framework.
- Basically, the configuration space of a robot is the set of all configurations and/or positions that the robot can attain.

Free Configuration Space and Configuration space Obstacle

This set of configurations that the robot cannot inhabit is referred to as a configuration space obstacle.



Multiple Obstacles (Minkowski sum of robot and obstacle)

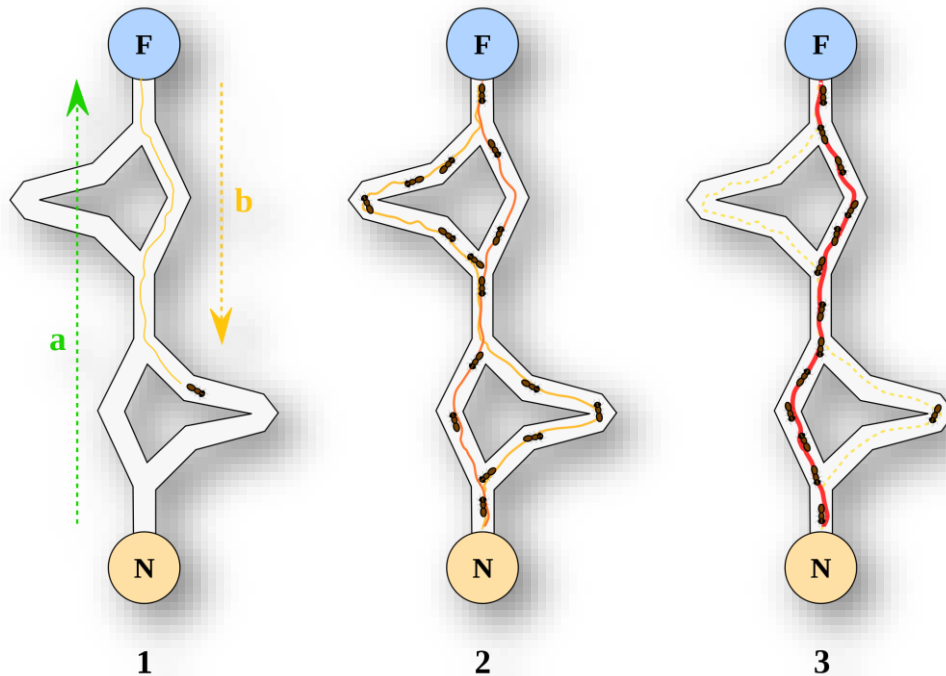


Metaheuristic Algorithms for Path planning

Ant Colony Optimization (ACO)

Particle Swarm Optimization PSO)

..... etc.



Manipulator (Industrial Robot) Trajectory

- An industrial robot to follow a pre-planned path is the largest problem of motion. This is termed as trajectory planning for the robot. During the trajectory planning, the object (or the work piece) has to be moved in accordance with a planned trajectory.
- The goal of the trajectory planning is to describe the requisite motion as a time sequence of joint-link end effector locations and their higher order derivatives. These are generated by interpolation or approximation by properly fitting a polynomial function.
- The trajectory planning serves as giving inputs or control set points to the robot's control system. The control system, in turn, assures that the robot executes the planned trajectories.

Manipulator (Industrial Robot) PTrajectory

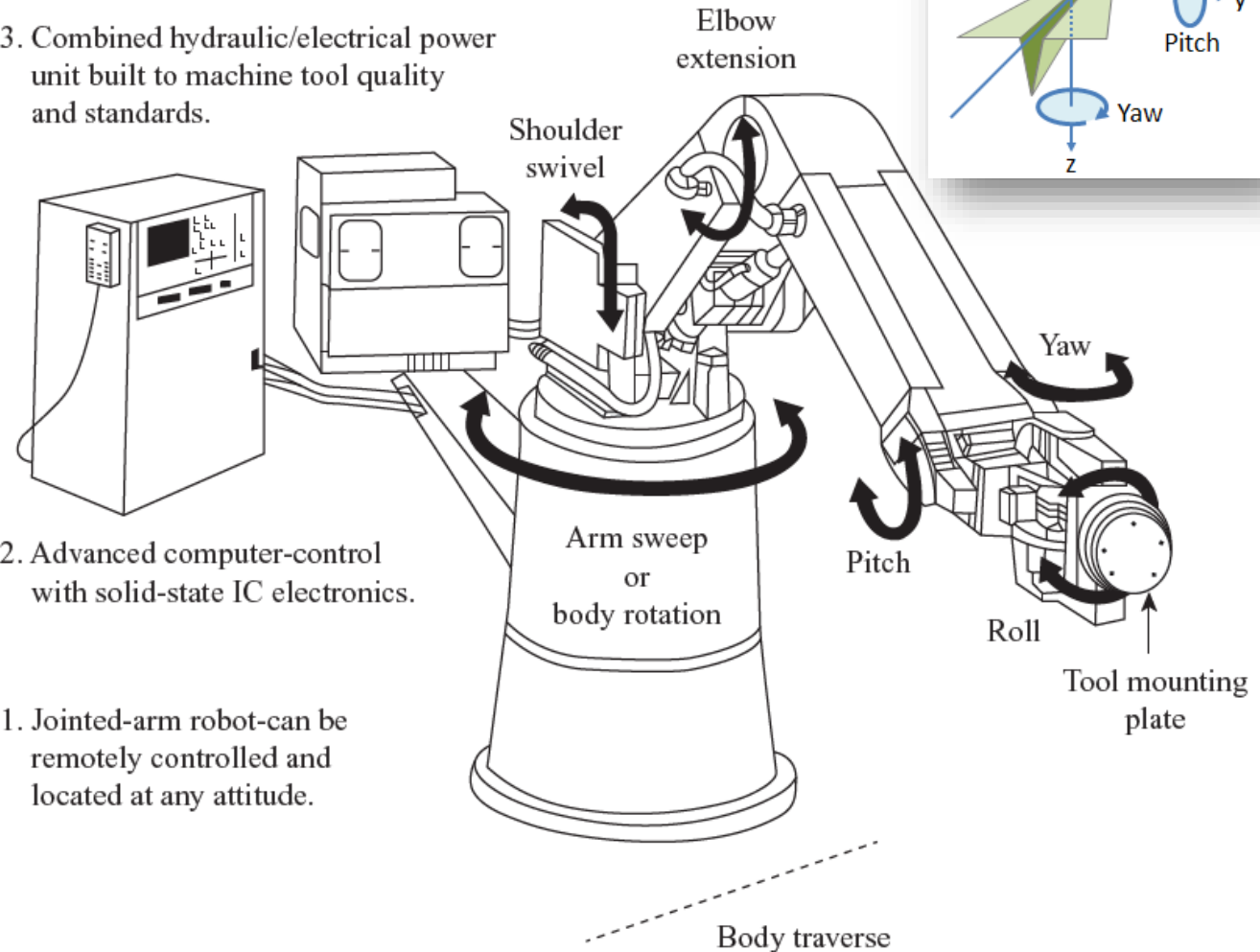


Cincinnati Milacron's T³

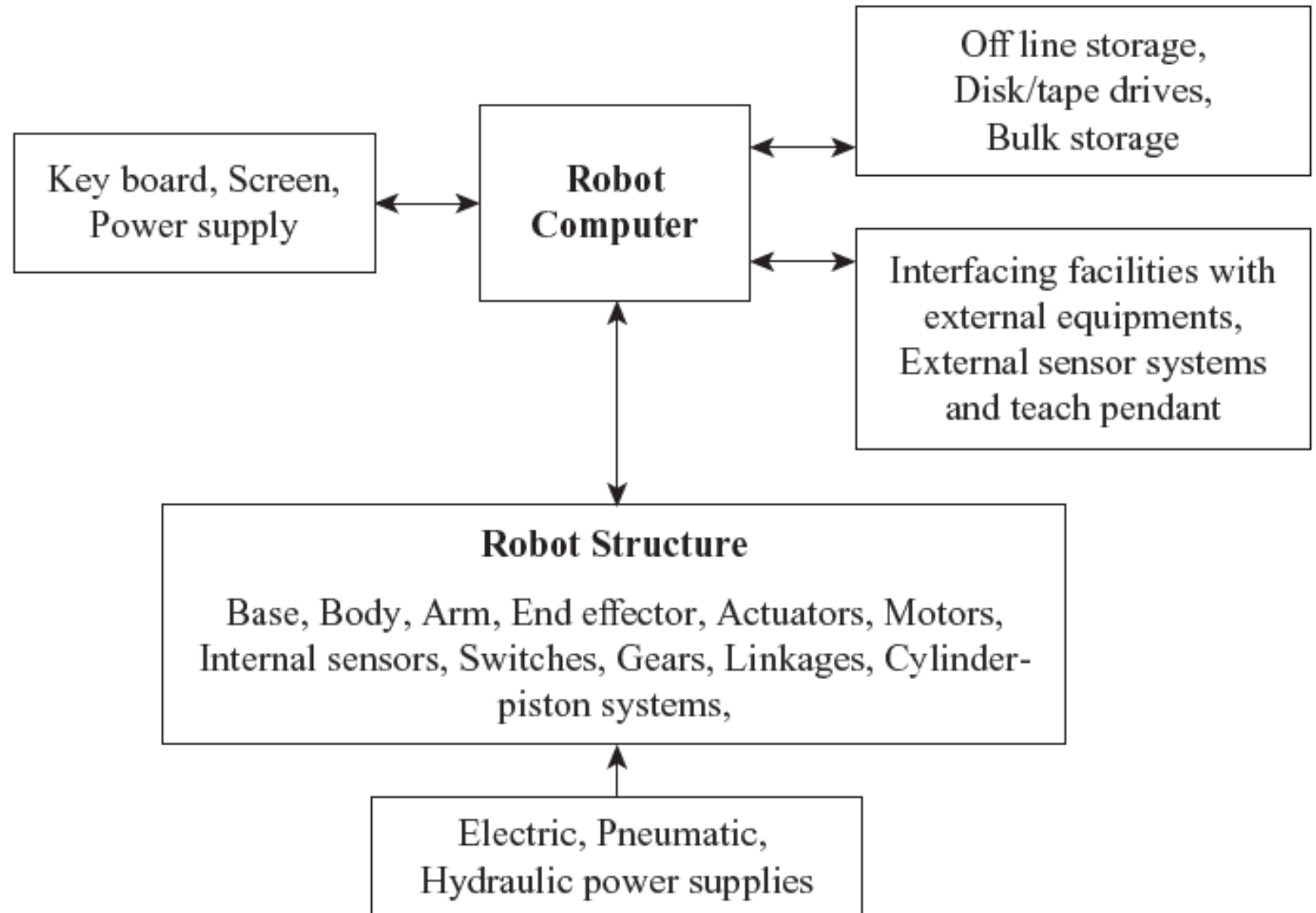
3. Combined hydraulic/electrical power unit built to machine tool quality and standards.

2. Advanced computer-control with solid-state IC electronics.

1. Jointed-arm robot-can be remotely controlled and located at any attitude.



Block Diagram of a Manipulator



Configurations of Industrial Robots



Articulated



SCARA



Spherical

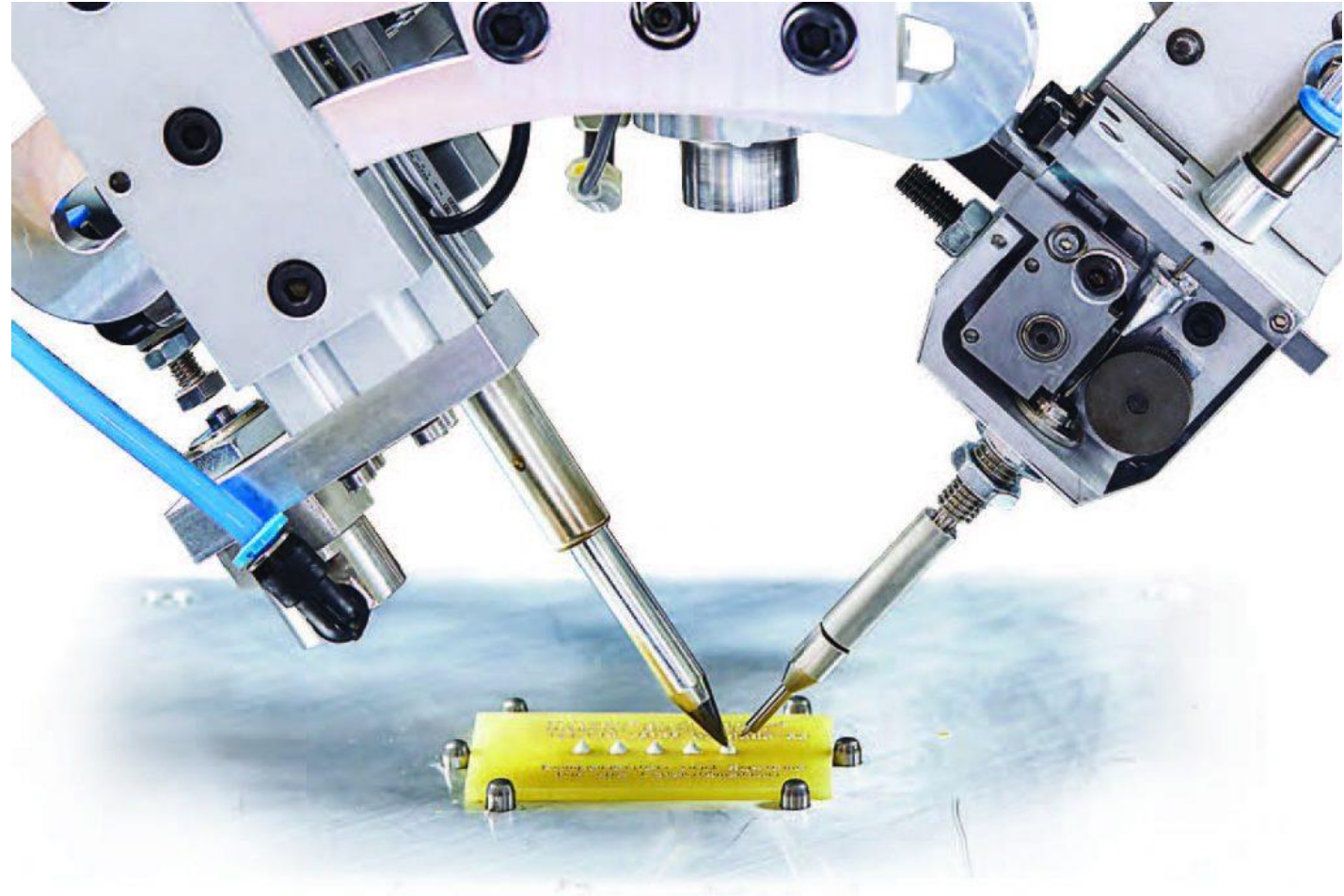


Cartesian

PRECISION OF MOVEMENT

Robot's capability to reach a point within its workspace (volume) depends on the precision of the robot. The precision of a robot is a function of three features:

- Spatial resolution
- Accuracy
- Repeatability



Spatial resolution

Spatial resolution of a robot is the smallest increment of movement in which the robot divides its entire workspace.

Example

Consider a robot having a sliding joint which can extend the link maximum to 1 m. The robot control memory has 12-bit word storage. What is the control resolution of the axis motion?

Solution:

Maximum number of increments within the joint range is $2^{12} = 4096$.

The minimum increment in sliding joint motion is $1 \text{ m} / 4096 = 0.24414 \text{ mm}$.

Therefore, the control resolution is 0.244 mm (to three digits accuracy).

Note:

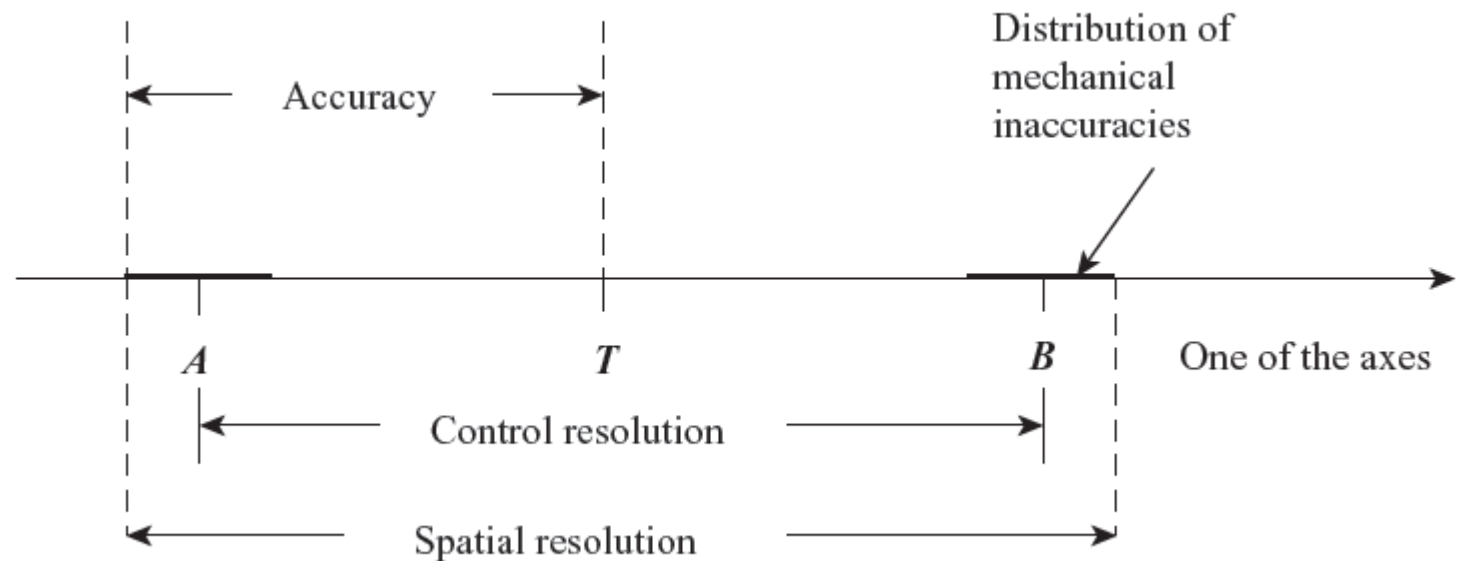
A robot has several joints, some of which are rotational and others translational. Each joint has its own joint range and hence its control resolution. Hence, the total control resolution of the entire robot cannot be obtained by algebraic addition of all individual resolutions. This is a complex procedure in which each resolution has to be considered as a vector in a three-dimensional space; the total resolution is, then, the vector addition of all joint resolutions.

Accuracy

Accuracy refers to a robot's ability to position its wrist end at a desired target point within its work volume. The accuracy is related to spatial resolution because the accuracy depends on how closely the neighboring points are defined within the work volume and on how accurately the mechanical structure is produced.

Example:

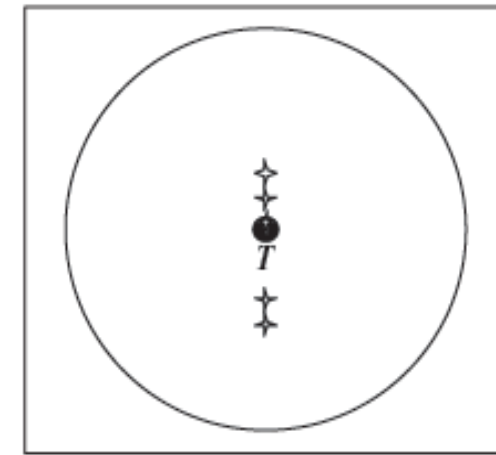
For the same example of Spatial resolution (previous slide), A and B are the nearest points as determined by control resolution.



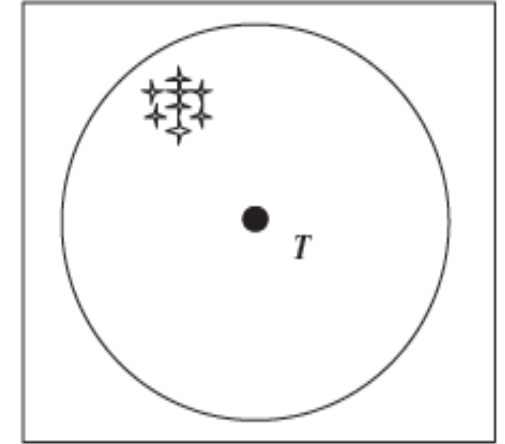
Repeatability



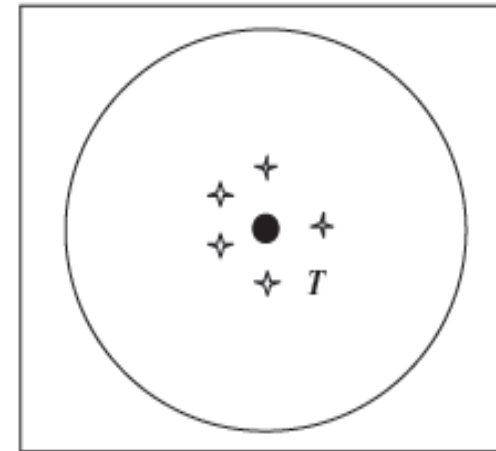
Accuracy and Repeatability: (a) Not Accurate Neither Repeatable, (b) Accurate But Not Repeatable, (c) Not Accurate But Repeatable and (d) Accurate and Repeatable



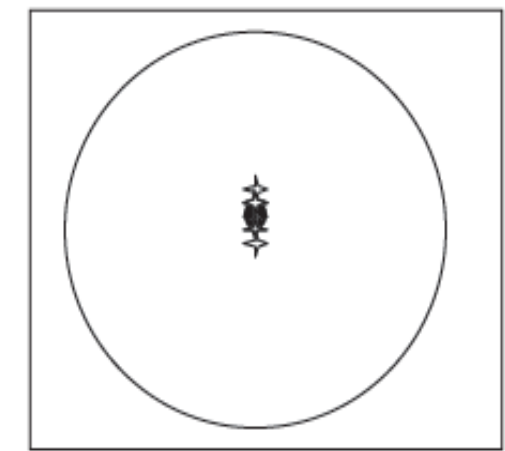
(a)



(c)



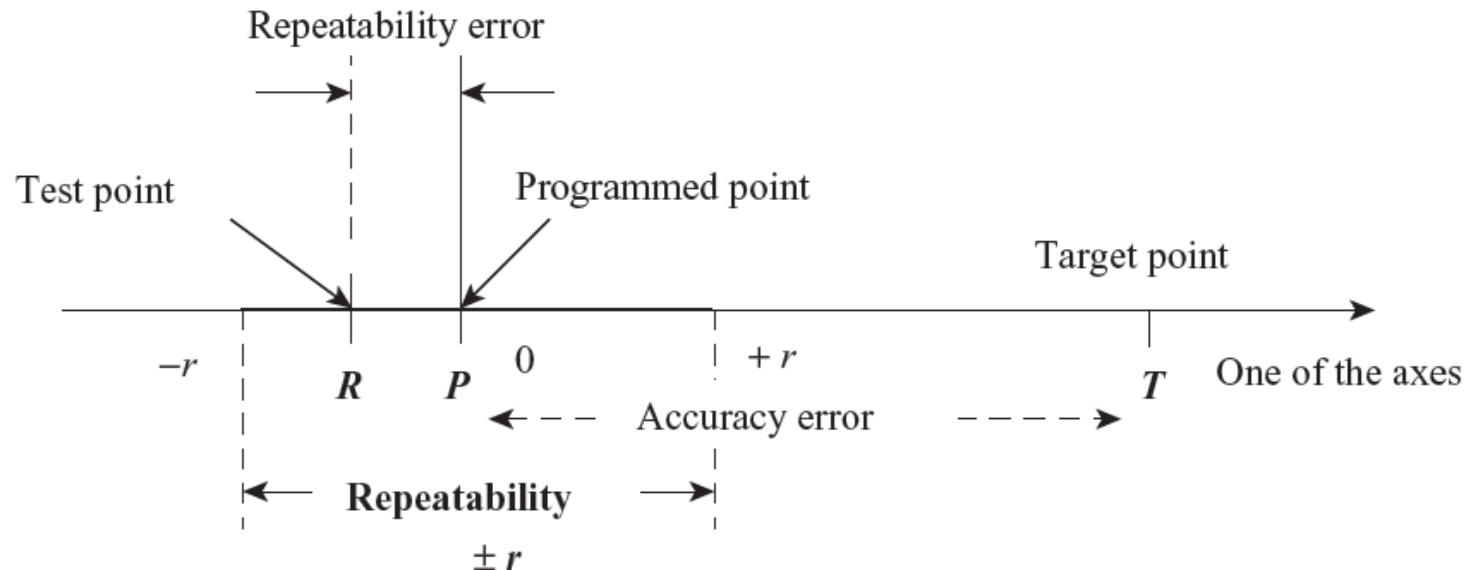
(b)



(d)

Measure of Repeatability

Assume a robot holds a pin at its gripper and is commanded to go to the target location on a board, kept horizontal, again and again and prick. By studying the test (pricked) points, we can estimate the robots ability of repeatability and accuracy. T is the location to which the robot is commanded to go. However, due to its errors and errors in control resolution, the robot will not reach T but it reaches the point P . If repeatability is the highest, the robot has to reach the point P whenever it is commanded to go to T . Due to its inability to reach point P over and over again, it goes to any other point along the bold short line around P . One such point is R which shows the repeatability error. The repeatability is now defined as $\pm r$ where $2r$ is the length of the bold short line which carries all the test points.



STEPS IN TRAJECTORY PLANNING

1. Task description: This is the first step in the motion planning problem. This task can be grouped into the following three different categories:

The first category is the pick and place application. This task is specified as initial and final end-effector locations. No particular specification on intermediate location of end effector is given and the planner is free to formulate any convenient path. The user specifies a goal point for an initial point, both points are known in Cartesian space.

In the second category, if in addition to start and finish points, a specific path is to be traced by the end effector in Cartesian space. This is known as continuous path motion and continuous trajectory. For example, in arc welding, the arc welder specifies the type and parameters of the path to be traced.

The third category of task description is where more than one set of points are specified. For example, a pick and place operation in a place where obstacles are present. The first point is called the initial point while the last point is called the goal point. The intermediate locations are via points. However, the task performed by tool or gripper in the point-to-point motion along the continuous path motion is not part of trajectory planning.

STEPS IN TRAJECTORY PLANNING, cont.

2. Employing a trajectory planning technique: Various trajectory planning techniques fall into one of the two categories - the joint space technique and Cartesian space technique.

In terms of point-to-point motion, with or without path points, cartesian space techniques may be employed. By adopting inverse kinematics, joint variables and their derivatives can be obtained.

3. Computing the trajectory: The final step is to compute the time sequence values and their derivatives attained by any function generated by trajectory planning techniques. These values are computed at a particular path update. The path update rate in real time lies between 20 and 200 Hz in a typical industrial robot system.

Modeling of Trajectory

A path (s) maps a scalar path parameter s , assumed to be 0 at the start of the path and 1 at the end, to a point in the robot's configuration space. As s increases from 0 to 1, the robot moves along the path. Sometimes s is taken to be time and is allowed to vary from time $s = 0$ to the total motion time $s = T$, a time scaling $s(t)$ assigns a value s to each time $t \in [0; T]$, $s : [0; T] \rightarrow [0; 1]$. Together, a path and a time scaling define a trajectory ($s(t)$), or (t) for short. Using the chain rule, the velocity and acceleration along the trajectory can be written as:

$$\dot{\theta} = \frac{d\theta}{ds} \dot{s}$$

$$\ddot{\theta} = \frac{d\theta}{ds} \ddot{s} + \frac{d^2\theta}{ds^2} \dot{s}^2$$

Point-to-point trajectory

A **straight line** from a start configuration θ_{start} to an end configuration θ_{end} could be defined in joint space or in task space. The advantage of a straight-line path from θ_{start} to θ_{end} in joint space is simplicity: since joint limits typically take the form $\theta_{i;\min} \leq \theta_i \leq \theta_{i;\max}$ for each joint i , the allowable joint configurations form a convex set Θ_{free} in joint space, so the straight line between any two endpoints in Θ_{free} also lies in Θ_{free} . The straight line can be written

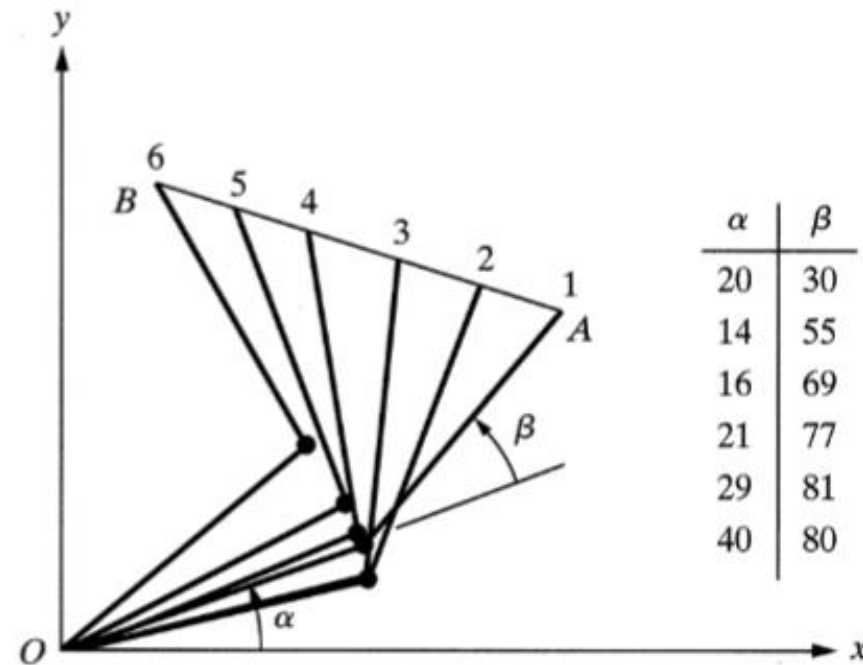
$$\theta(s) = \theta_{\text{start}} + s(\theta_{\text{end}} - \theta_{\text{start}}), \quad s \in [0, 1]$$

with derivatives

$$\begin{aligned} \frac{d\theta}{ds} &= \theta_{\text{end}} - \theta_{\text{start}}, \\ \frac{d^2\theta}{ds^2} &= 0. \end{aligned}$$

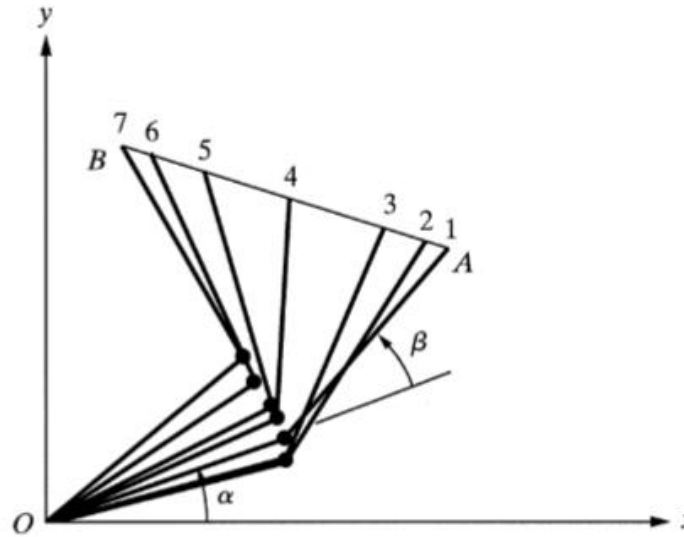
Straight Line Trajectory- Version A

- ◆ Let us assume that the robot gripper follows a straight line between points A and B.
- ◆ The simplest way is to draw a line (interpolate) between A, B.
- ◆ Divide the line into five segments and solve for necessary angles α and β at each point.
- ◆ The joint angles do not change uniformly.



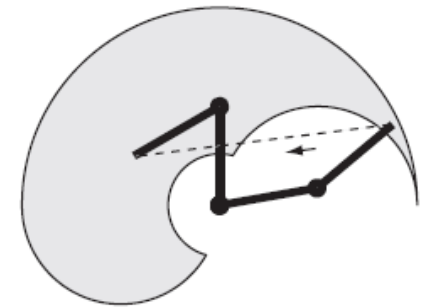
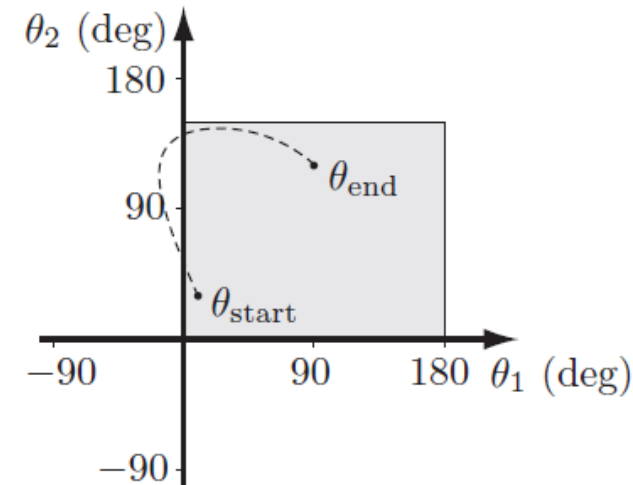
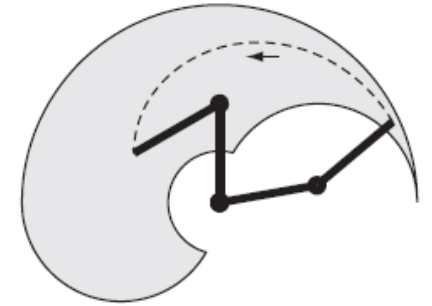
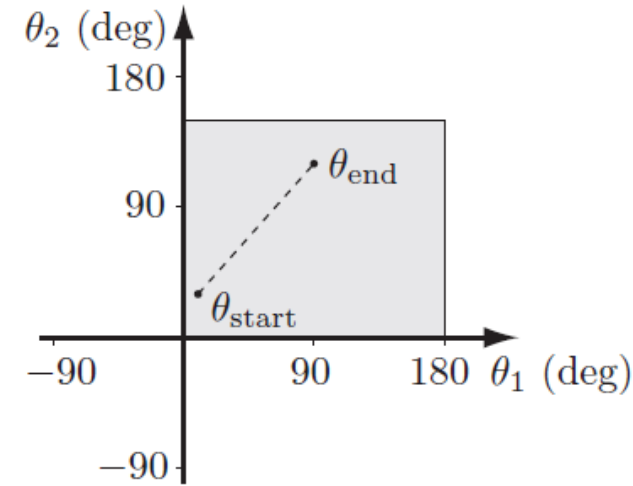
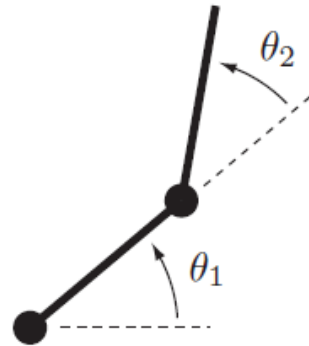
Straight Line Trajectory- Version B

- ◆ Again interpolation between A, B by a straight line.
- ◆ The aim is to accelerate at the beginning and decelerate at the end.
- ◆ Divide the segments differently.
 - The arm moves at smaller segments as we speed up at the beginning;
 - Go at a constant cruising rate;
 - Decelerate with smaller segments as approaching Point B.



REMARK

- Straight lines in joint space generally do not yield straight-line motion of the end-effector in task space. If task-space straight-line motions are desired, the start and end configurations can be specified by X_{start} and X_{end} in task space.
- Since the robot's reachable task space may not be convex in X coordinates, some points on a straight line between two reachable endpoints may not be reachable.
- be reachable.



$$0^\circ \leq \theta_1 \leq 180^\circ, 0^\circ \leq \theta_2 \leq 150^\circ$$

Smoothness

We want position and orientation to vary smoothly with time

- reduce the peak acceleration of the robot
- Reduce the size of motor required
- Reduce vibration in the structure of the robot

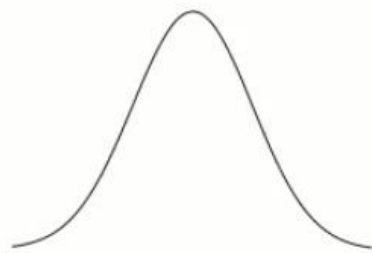
What is smooth?

- Position of the robot $x(t)$ is continuous
- Velocity of the robot $\dot{x}(t)$ is continuous
- Acceleration of the robot $\ddot{x}(t)$ is continuous
- Jerk of the robot $\dddot{x}(t)$ is continuous

Examples of Smooth Functions



sinusoid



Gaussian



polynomial

Polynomial Trajectory (Time Scaling) LINEAR FUNCTION WITH PARABOLIC BLENDS (LFPB)

We shall study the use of a p -degree ($p = 3$) polynomial as the interpolation function when a set of path points is given. For a smooth motion between two points, the selection of a single polynomial for entire joint path depends on the number of constraints such as (i) initial position, (ii) initial velocity, (iii) final position and (iv) final velocity.

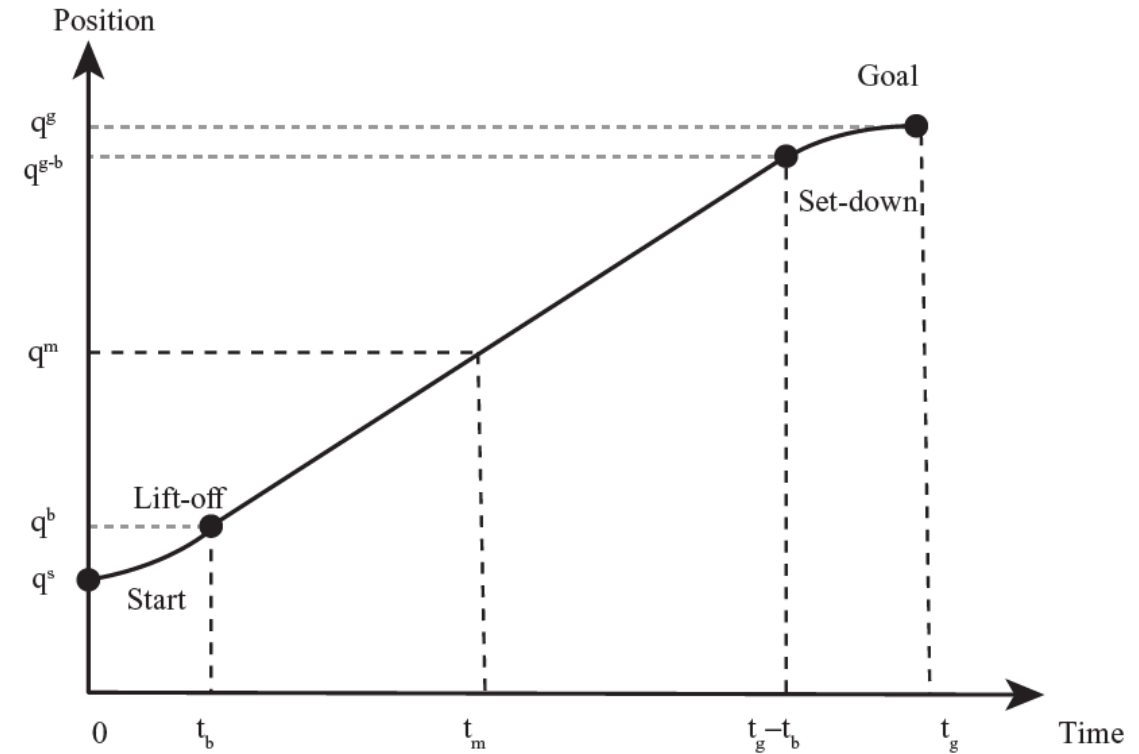
A third degree ($p = 3$) polynomial (cubic spline) with four coefficients can be used as:

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3;$$

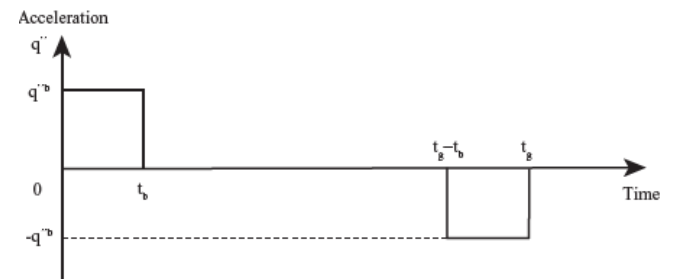
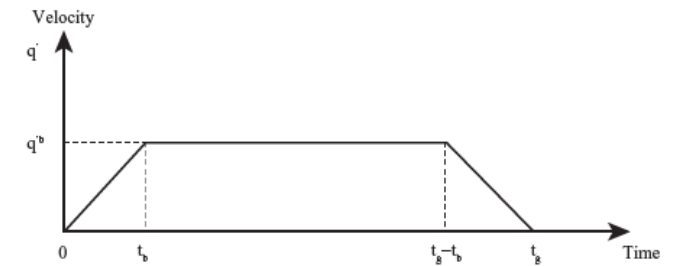
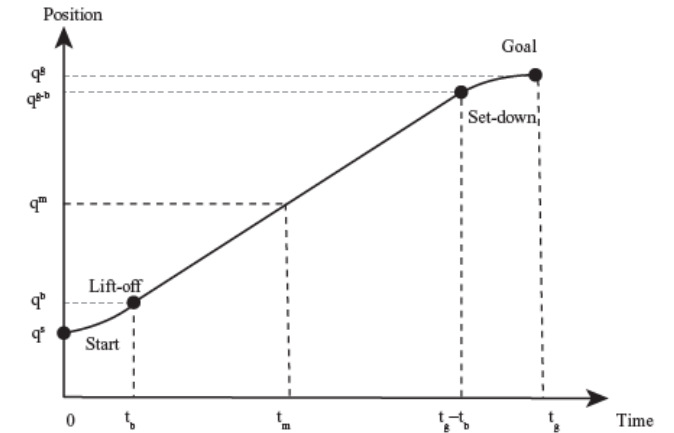
In a typical pick and place operation, the robot begins moving at constant acceleration, away from initial position in an appropriate direction. For example, in a vertically upward direction from a horizontal surface, the initial position is complete. After it has moved to a safe distance, lift-off is complete. From then onwards, it moves at a constant velocity with zero acceleration.

This constant velocity and zero acceleration motion continue until it reaches the set-down position. From this point it begins to vertically decelerate reaching the final position.

This is known as linear function with parabolic blends (LFPB) and has six constraints - four initial and final position and velocity respectively, one acceleration constraint in the beginning and one deceleration constraint in the final stage.



Position, Velocity and Acceleration Profiles



Time Scaling by Third-order Polynomials

A convenient form for the time scaling $s(t)$ is a cubic polynomial of time,

$$s(t) = a_0 + a_1t + a_2t^2 + a_3t^3.$$

A point-to-point motion in time T imposes the initial constraints $s(0) = d(s)/d(t)(0) = 0$ and the terminal constraints $s(T) = 1$ and $d(s)/d(t)(T) = 0$.

Hence,

$$\dot{s}(t) = a_1 + 2a_2t + 3a_3t^2$$

at $t = 0$ and $t = T$ and solving the four constraints for $a_0; \dots; a_3$, we find

$$a_0 = 0, \quad a_1 = 0, \quad a_2 = \frac{3}{T^2}, \quad a_3 = -\frac{2}{T^3}.$$

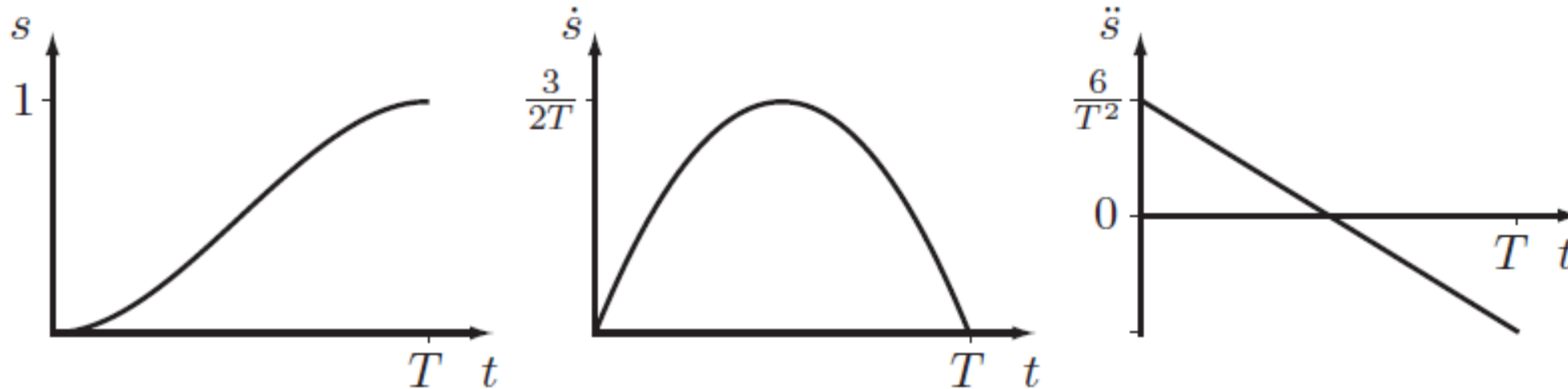
Substituting $s = a_2 t^2 + a_3 t^3$ into Equation $\theta(s) = \theta_{\text{start}} + s(\theta_{\text{end}} - \theta_{\text{start}})$ yields

$$\theta(t) = \theta_{\text{start}} + \left(\frac{3t^2}{T^2} - \frac{2t^3}{T^3} \right) (\theta_{\text{end}} - \theta_{\text{start}}),$$

$$\dot{\theta}(t) = \left(\frac{6t}{T^2} - \frac{6t^2}{T^3} \right) (\theta_{\text{end}} - \theta_{\text{start}}),$$

$$\ddot{\theta}(t) = \left(\frac{6}{T^2} - \frac{12t}{T^3} \right) (\theta_{\text{end}} - \theta_{\text{start}}).$$

Position, Velocity and Acceleration Profiles

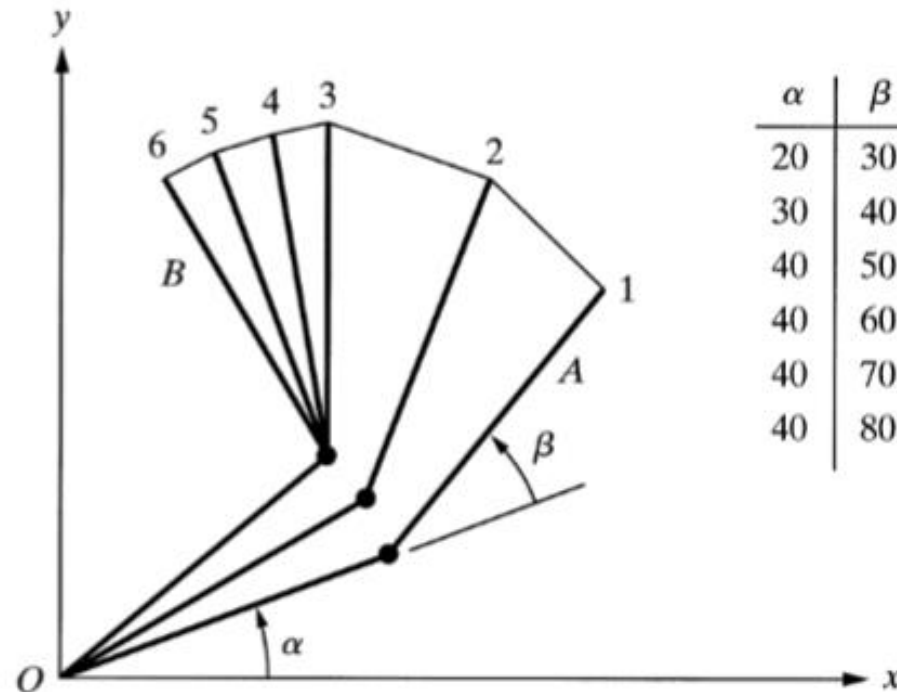


Note:

- The maximum joint velocities are achieved at the halfway point of the motion, $t = T/2$.
- The maximum joint acceleration and deceleration are achieved at $t = 0$ and $t = T$.

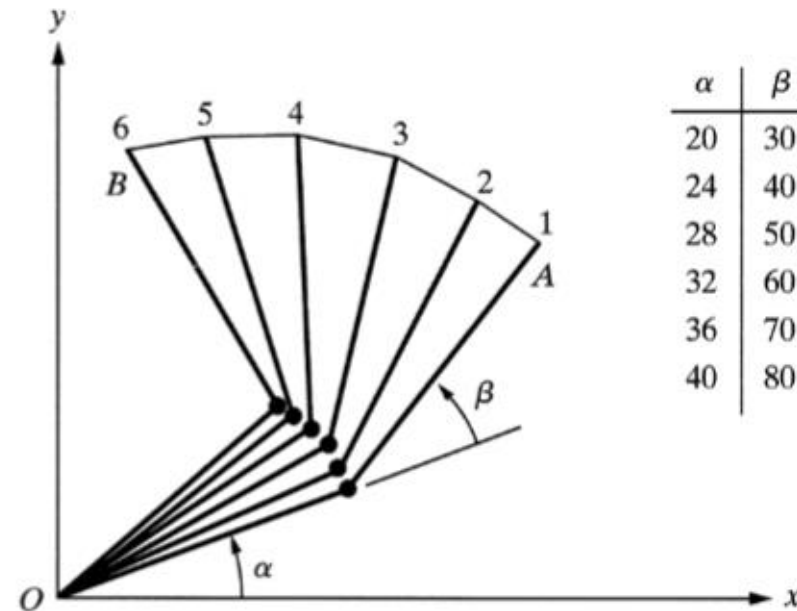
Time scaling: Non-Normalized trajectory

- ◆ Move the robot from A to B, to run both joints at their maximum angular velocities.
- ◆ After 2 [sec], the lower link will have finished its motion, while the upper link continues for another 3 [sec]
- ◆ The path is irregular and the distances traveled by the robot end point are not uniform.



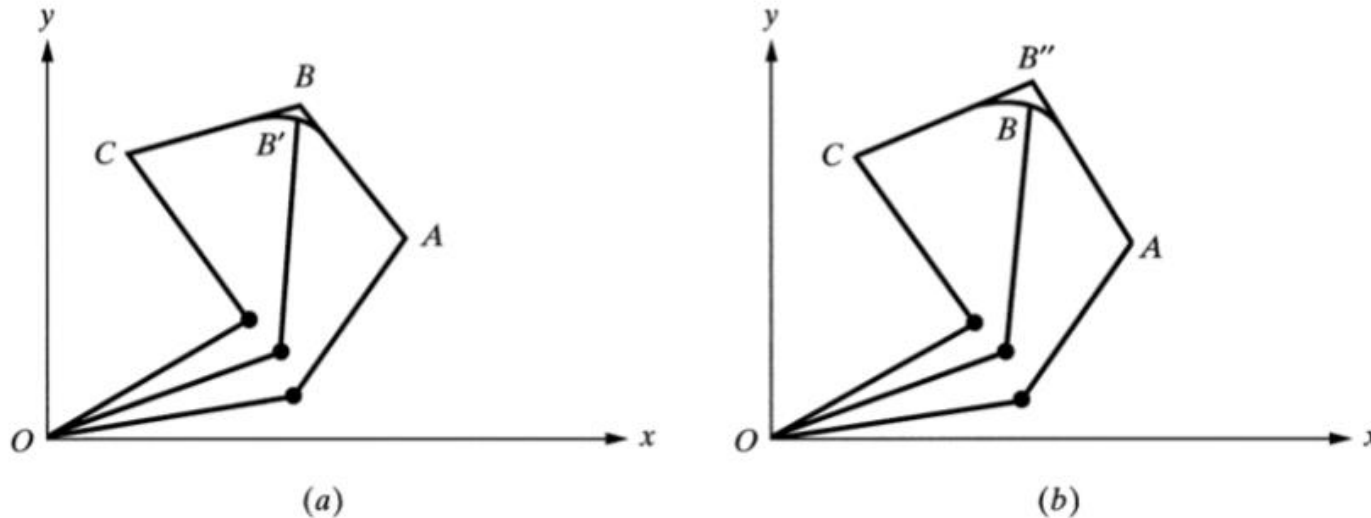
Time Scaling: Normalized trajectory

- ◆ Let us assume that the motions of both joints are normalized by a common factor such that the joint with smaller motion will move proportionally slower and the both joints will start and stop their motion simultaneously.
- ◆ Both joints move at different speeds, but move continuously together.
- ◆ The resulting trajectory will be different.



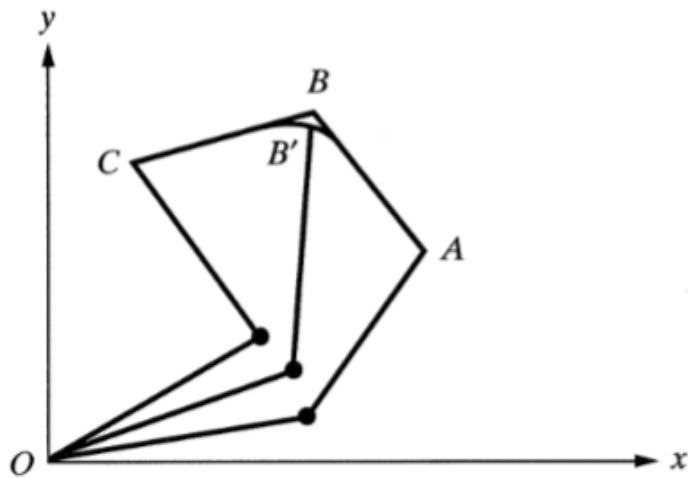
Manipulator Path Planning: Continuous Transition , Version A

- ◆ Stop-and-go motion through the way-point list creates jerky motions with unnecessary stops.
- ◆ How? Blend the two portions of the motion at Point B.

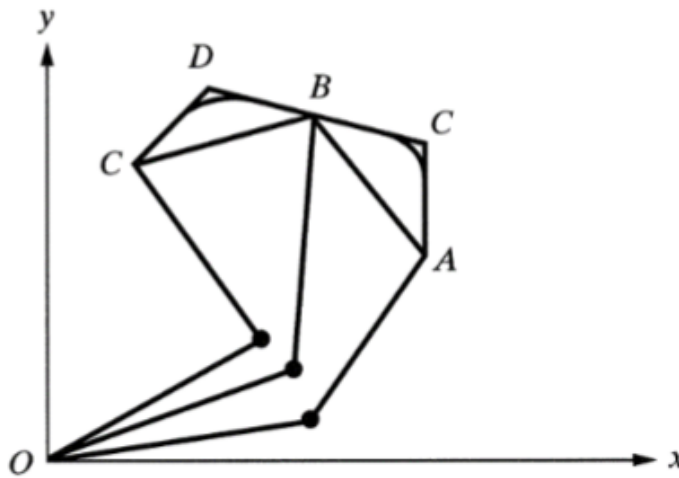


Manipulator Path Planning: Continuous Transition , Version B

- ◆ Alternative scheme ensuring that the trajectory passes through control points.
- ◆ Two way-points D and E are picked such that Point B will fall on the straight-line section of the segment ensuring that the robot will pass through Point B.



(a)



(b)

Example:

The SCARA robot has four joints and four limbs. Its second joint is required to move from 30° to 150° in 5 s. Determine the cubic polynomial trajectory to generate smooth trajectory for the joint. What are the maximum velocity and acceleration for this trajectory?

Solution:

The given parameters of the trajectory are as follows:

$$\begin{aligned}\theta^s &= 30^\circ, & \theta^g &= 150^\circ, \\ v^s &= 0, & v^g &= 0, \\ t^0 &= 0 \text{ s}, & t^g &= 5 \text{ s}.\end{aligned}$$

Solution:

The four coefficients of a cubic polynomial are computed as follows:

$$a_0 = 30,$$

$$a_1 = 0,$$

$$a_2 = (3/t_g^2) (\theta^g - \theta^s) = 14.4,$$

$$a_3 = -(2/t_g^3) (\theta^g - \theta^s) = -1.92.$$

Thus, we get the following polynomials interpolating position, velocity and acceleration profiles:

$$\theta(t) = 30.0 + 14.4 t^2 - 1.92 t^3,$$

$$\theta'(t) = 28.8 t - 5.76 t^2,$$

$$\theta''(t) = 28.8 - 11.52 t.$$

The velocity profile is parabolic and acceleration profile is linear. The maximum values of velocity and acceleration are $2.5^\circ/\text{s}$ and $-11.52^\circ/\text{s}^2$, respectively.

Example:

A single cubic trajectory given by $q(t) = 30 + t^2 - 6t^3$

is used for a period of 3 s. Determine the start and goal positions, velocity and acceleration of the end effector.

Solution:

Differentiating the function twice, we get

$$q'(t) = 2t - 18t^2 \qquad q''(t) = 2 - 36t.$$

Thus, at $t = 0$, start and goal positions are

$$t = 0; q(0) = 30,$$

$$t = 3; q(3) = -123.$$

Velocity and acceleration are

$$q'(3) = -156$$

$$q''(3) = -106$$

Reading List

1. Ramachandran N., Introduction to Industrial Robotics, Pearson Publishers, 2016.
2. Kevin M. Lynch and Frank C. Park, Modern Robotics, Cambridge Uni Press, 2017.

Questions?